# Commutative Algebra and Algebraic Geometry

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**Note:**
Due to the mismatch in concepts taught in the lecture and recitation, some material is mentioned in this summary before it is formally introduced. For example, the spectrum of a ring is discussed in the context of integral algebras, before it is formally defined in its own section. It is advised to read the section on ring spectra early on (before going in depth on commutative algebra).

# 1 Commutative Algebra

## 1.1 Rings and Ideals

We recall some basic notions regarding rings.

**1.1.1 Definition**
A **ring** is a set $R$ equipped with two operations $+$ and $\cdot$, making it into a commutative group and a monoid respectively. Furthermore, they satisfy the distributivity properties:

$$a \cdot (b + c) = ab + ac, \qquad (a + b) \cdot c = ac + bc$$

for all $a, b, c \in R$. The additive and multiplicative identities are denoted 0 and 1 respectively.

**Note:**
In this course all rings are assumed to be **commutative**: $\cdot$ forms a commutative monoid.

**1.1.2 Definition**
A **ring morphism** is a function between two rings, $f\colon R \to S$, which preserves structure: $f(x + y) = f(x) + f(y)$, $f(xy) = f(x)f(y)$, and $f(1) = 1$. A **ring isomorphism** is a ring morphism with an inverse (equivalently a bijective ring morphism).
A **subring** of a ring $R$ is a subset $S \subseteq R$ equipped with a ring structure making the inclusion $S \to R$ a ring morphism.
An **ideal** of a ring $R$ is a nonempty subset $I \subseteq R$ closed under addition and multiplication by $R$: for $r \in R$ and $x \in I$, $rx \in I$. For an ideal $I \subseteq R$, we define its **quotient ring** $R/I$ to be the set of all cosets $[r] = r + I$ with ring structure given by

$$0 = [0] = I, \qquad 1 = [1], \qquad [x] + [y] = [x + y], \qquad [x] \cdot [y] = [xy],$$

equivalently the unique structure making the projection map $\pi\colon R \to R/I, r \mapsto [r]$ a ring morphism.

**1.1.3 Proposition**
Let $f\colon R \to S$ be a ring morphism, then

**(1)** For every ideal $J \leq S$, $f^{-1}(J) \leq R$ is an ideal.

**(2)** In particular $f$'s **kernel**, $\ker f = f^{-1}0$, is an ideal of $R$. And $f$'s **image**, $\mathrm{im} f = f(R)$ is a subring of $S$. $f$ then quotients to give a ring isomorphism $R/\ker f \cong \mathrm{im} f$.

**(3)** For an ideal $I \leq R$, consider the projection $\pi\colon R \to R/I$. Then $J \mapsto \overline{J} = \pi(J)$ and $\overline{J} \mapsto \pi^{-1}\overline{J}$ gives a one-to-one correspondence between ideals of $R$ containing $I$ and ideals of $R/I$.

**1.1.4 Definition**

For a ring $R$ and a subset $S \subseteq R$, we denote the smallest ideal containing $S$ by $(S)$. This exists as the arbitrary intersection of ideals is an ideal. When $S = \{x\}$ is a singleton, we denote $(\{x\}) = (x)$; such an ideal is called a **principal ideal**.

**1.1.5 Proposition**

The ideal generated by $S$, $(S)$, is the set of all finite $R$-combinations of elements from $S$:

$$(S) = \left\{ \sum_{s \in S} r_s s \;\middle|\; r_s \in R, \text{ all but finitely many being zero} \right\}.$$

In particular, $(x) = Rx$ is the set of all multiples of $x$.

**Proof**

It is not hard to see that this is an ideal, and clearly every ideal containing $S$ must contain it. ■

**1.1.6 Definition**

Let $R$ be a ring, an element $x \in R$ is

**(1)** a **zero divisor** if there is a $y \in R$ such that $xy = 0$;

**(2)** a **unit** if there is a $y \in R$ such that $xy = 1$;

**(3)** **nilpotent** if $x^n = 1$ for some $n > 0$.

The set of units of $R$ is denoted $R^\times$ and forms a group.

**1.1.7 Definition**

A ring $R \neq 0$ is

**(1)** **reduced** if every $x \neq 0$ is not nilpotent;

**(2)** an **integral domain** if every $x \neq 0$ is not a zero divisor;

**(3)** a **field** if every $x \neq 0$ is a unit.

It is easy to see that a field is an integral domain is reduced.

**1.1.8 Proposition**

For a ring $R$, the following are equivalent:

**(1)** $R$ is a field,

**(2)** $R$ has no non-trivial ideals,

**(3)** Every ring morphism out of $R$ to a non-zero ring is injective.

**Proof**

If $R$ is a field, then every nonzero element is a unit and thus the ideal generating it must be all of $R$. Conversely if $R$ has no non-trivial ideals, then for $x \neq 0$ we must have that $1 \in (x)$ and thus there is a $y \in R$ such that $xy = 1$ so it is a unit.

If $R$ has no non-trivial ideals then for every $f: R \to S \neq 0$ we must have $\ker f = 0$, as it is an ideal not containing 1. Thus $f$ is injective. Conversely if $I \neq R$ is an ideal of $R$, then $\pi: R \to R/I$ is injective, but its kernel is $I$, so $I = 0$. ■

**1.1.9 Proposition**

If $\{I_\alpha\}_\alpha$ is a collection of ideals from $R$, then

**(1)** $\bigcap_\alpha I_\alpha$ is an ideal, the largest ideal contained in each $I_\alpha$.

**(2)** $\sum_\alpha I_\alpha = \{\sum_\alpha x_\alpha \mid x_\alpha \in I_\alpha \text{ all but finitely many are zero}\}$ is an ideal, the smallest ideal containing each $I_\alpha$.

**Proof**

The first point is easily verified, and the second one follows because

$$\sum_\alpha I_\alpha = \left(\bigcup_\alpha I_\alpha\right).$$

■

**1.1.10 Definition**

If $I, J$ are ideals then we define their **product** to be the ideal generated from all products of elements from $I$ and $J$:

$$IJ = (\{xy \mid x \in I, y \in J\}).$$

Since $I, J$ are ideals and are thus closed under multiplication from $R$, for $x \in I$ and $y \in J$ we have that $xy \in I \cap J$. Thus $IJ \subseteq I \cap J$.

**1.1.11 Definition**

For an ideal $I \subseteq R$, we define its **radical** to be

$$\sqrt{I} = \{r \in R \mid r^n \in I \text{ for some } n \geq 0\}.$$

**1.1.12 Lemma**

The radical of an ideal is an ideal.

**Proof**

If $a^n \in I$ and $r \in R$ then $(ra)^n = r^n a^n \in I$ so that $ra \in \sqrt{I}$. And if $a^n, b^m \in I$ then

$$(a+b)^{n+m} = \sum_{i=0}^{n+m} \binom{n+m}{i} a^i b^{n+m-i} \in I,$$

since for each $i$, if $i < n$ then $n+m-i > m$, so $a+b \in \sqrt{I}$. ■

## 1.2 Prime and maximal ideals

**1.2.1 Definition**

A proper ideal $I \subset R$ is

**(1)** **prime** if $ab \in I$ implies $a \in I$ or $b \in I$. Equivalently if $a, b \notin I$ then $ab \notin I$.

**(2)** **maximal** if it is not contained in any other proper ideal.

**1.2.2 Proposition**

An ideal $I$ is

**(1)** prime iff $R/I$ is an integral domain;

**(2)** maximal iff $R/I$ is a field.

Consequently, a maximal ideal is prime.

**Proof**

$R/I$ is an integral domain iff $[a][b] \neq [0]$ when $[a], [b] \neq [0]$. This means that $ab \notin I$ when $a, b \notin I$, as desired. $R/I$ is a field iff it has no nontrivial ideals. Since ideals of $R/I$ are in bijection with ideals containing $I$, this is iff $I$ is maximal. ■

In particular the zero ideal $0 = (0) = \{0\}$ is prime iff $R \cong R/(0)$ is an integral domain.

**1.2.3 Corollary**

Let $I$ be an ideal, then the correspondence $J \leftrightarrow \overline{J}$ between ideals containing $I$ and ideals of $R/I$ preserves primality and maximality: $J$ is prime/maximal iff $\overline{J}$ is.

**Proof**

We have that

$$(R/I)/\overline{J} = (R/I)/(J/I) \cong R/J,$$

so that $(R/I)/\overline{J}$ is an integral domain/field iff $R/J$ is. The result follows. ■

**1.2.4 Definition**

**(1)** An element $0 \neq x \in R$ is **irreducible** if it is not a unit, and if $x = yz$ then either $y$ or $z$ is a unit.

**(2)** An integral domain $R$ is a **principal ideal domain** (or a **PID**) if all of its ideals are principal.

**1.2.5 Proposition**

If $R$ is a PID and $0 \neq x$, then $(x)$ is prime iff $x$ is irreducible iff $(x)$ is maximal.

**Proof**

If $(x)$ is prime, $x$ is not a unit since it is a proper ideal. And if $x = yz$ we have that $y$ or $z$ is in $(x)$, say $y = xt$. Then $x = yz = xtz$, and so $x(1 - tz) = 0$, and since $R$ is an integral domain this means $tz = 1$ so $z$ is a unit.

Now if $x$ is irreducible, then suppose $(x) \subseteq I$ is a proper ideal containing $(x)$. Since $R$ is a PID, $I = (y)$ so that $x \in (y)$ meaning $x = yz$. $(y)$ is a proper ideal so $y$ is not a unit, thus $z$ is, and then $y = xz^{-1}$ so that $y \in (x)$ and thus $(x) = (y)$. So $(x)$ is maximal, as desired. ■

**1.2.6 Example**

The ring of integers $\mathbb{Z}$ is a PID. Its irreducible elements are precisely its prime elements, so that its prime ideals are $(0)$ and $(p)$ for $p \in \mathbb{Z}$ prime.

**1.2.7 Example**

If $k$ is a field, then its ring of polynomials $k[x]$ is a PID. Indeed, if $0 \neq I \leq k[x]$ is an ideal, then let $f \in I$ have minimal degree. By polynomial division we see that $I = (f)$: for $g \in I$ we have that $g = fq + r$, but since $r = g - fq \in I$ has smaller degree than $f$, it must be zero.

Its prime ideals are $(0)$ and $(f)$ for an irreducible monic polynomial $f$.

**1.2.8 Lemma**

If $f\colon R \to S$ is a ring morphism, and $\mathfrak{q} \leq S$ is prime, then $f^{-1}\mathfrak{q} \leq R$ is prime.

**Proof**

A direct proof is easy, instead consider the composition:

$$\overline{f}\colon R \xrightarrow{f} S \xrightarrow{\pi} S/\mathfrak{q}.$$

By definition $f^{-1}\mathfrak{q} = \ker\overline{f}$, and so $\overline{f}$ induces an embedding $R/f^{-1}\mathfrak{q} \hookrightarrow S/\mathfrak{q}$. Subrings of integral domains are integral domains, so $R/f^{-1}\mathfrak{q}$ is an integral domain, thus $f^{-1}\mathfrak{q}$ is prime. ■

Subrings of fields are not fields, so the above proof does not show that the preimage of a maximal ideal is maximal. Indeed, this is not true: consider the unique map $f\colon \mathbb{Z} \to \mathbb{Q}$. Then $0 \leq \mathbb{Q}$ is maximal, but $f^{-1}0 = \ker f = 0$ is not maximal in $\mathbb{Z}$.

**1.2.9 Theorem**

For every ring $R$ and every proper ideal $I < R$, there is a maximal ideal $\mathfrak{m}$ containing $I$.

**Proof**

Consider the set of all proper ideals containing $I$. This is nonm-empty, as it contains $I$. Furthermore, the union of every chain of proper ideals is a proper ideal; it being an ideal is easy to see, and it is proper since 1 is not in any element of the chain. Thus by Zorn, this set has a maximal element, which must be a maximal ideal. ■

**1.2.10 Corollary**

Let $x \in R$, then $x$ is a unit iff $x$ is not in any maximal ideal. That is,

$$R^\times = \bigcap_{\mathfrak{m}} \mathfrak{m}^c,$$

where the intersection is over all maximal ideals of $R$.

**Proof**

An element is a unit iff it is not contained in any proper ideal. This is clear enough: if $x$ is not a unit then $(x)$ is proper, and proper ideals don't contain units. Every proper ideal can be extended to a maximal ideal, so the result follows. ■

**1.2.11 Definition**

Let $R$ be a ring. Then define its **nilradical** to be $\mathrm{nil}(R) = \sqrt{0}$, i.e. the ideal of all nilpotent elements.

Clearly $R$ is reduced iff $\mathrm{nil}(R) = 0$.

**1.2.12 Proposition**

Let $I \leq R$ be an ideal, and let $\pi\colon R \to R/I$ be the projection. Then $\sqrt{I} = \pi^{-1}\mathrm{nil}(R/I)$.

**Proof**

We note that if $x^n \in I$ then $\pi(x)^n \in I$ so that $\pi(x) \in \mathrm{nil}(R/I)$ as desired. Conversely, if $\pi(x) \in \mathrm{nil}(R/I)$ then $\pi(x^n) = I$ so that $x^n \in I$ as desired. ■

**1.2.13 Proposition**

For a ring $R$,

**(1)** $R/\mathrm{nil}(R)$ is reduced;

**(2)** The nilradical is the intersection of all prime ideals:

$$\mathrm{nil}(R) = \bigcap_{\mathfrak{p}} \mathfrak{p}.$$

**Proof**

We noted above that $\mathrm{nil}R = \sqrt{\mathrm{nil}R} = \pi^{-1}\mathrm{nil}(R/\mathrm{nil}(R))$. But we have also that $\mathrm{nil}R = \pi^{-1}0$, so $\pi^{-1}0 = \pi^{-1}\mathrm{nil}(R/\mathrm{nil}(R))$ and since $\pi$ is surjective this means $\mathrm{nil}(R/\mathrm{nil}(R)) = 0$ so that $R/\mathrm{nil}(R)$ is reduced.

Clearly if $x \in \mathrm{nil}(R)$ then $x \in \mathfrak{p}$ for every prime $\mathfrak{p}$ since $x^n = 0 \in \mathfrak{p}$. Now if $x \notin \mathrm{nil}(R)$ then we claim there exists a prime ideal $\mathfrak{p}$ such that $x \notin \mathfrak{p}$. We prove this using Zorn. Let $T = \{1, x, x^2, \ldots\}$ and let $S$ be the poset of all ideals $I$ such that $I \cap T = \emptyset$; note that all ideals in $S$ are proper. As explained before the union of a chain of ideals is an ideal, and clearly if all of the ideals in the union don't intersect $T$, neither does the union. Thus by Zorn, $S$ has a maximal element $\mathfrak{p}$.

We claim that $\mathfrak{p}$ is prime. Indeed, suppose that $ab \in \mathfrak{p}$ but $a, b \notin \mathfrak{p}$. Then by maximality we have that $x^n \in (a, \mathfrak{p})$ and $x^m \in (b, \mathfrak{p})$ for some $n, m \geq 0$. But then $x^{n+m} \in (a, \mathfrak{p})(b, \mathfrak{p}) \subseteq \mathfrak{p}$, a contradiction.

Thus there is a prime ideal $\mathfrak{p}$ such that $\mathfrak{p} \cap T = \emptyset$, in particular $x \notin \mathfrak{p}$. So if $x \notin \mathrm{nil}(R)$ then $x \notin \bigcap_{\mathfrak{p}} \mathfrak{p}$, as desired. ■

**1.2.14 Corollary**

Let $I$ be an ideal of $R$, then

$$\sqrt{I} = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p},$$

where the intersection is over all prime ideals containing $I$.

**Proof**

We showed above that $\sqrt{I} = \pi^{-1}\mathrm{nil}(R/I)$. By above $\mathrm{nil}(R/I)$ is the intersection over all prime ideals of $R/I$, which are all prime ideals of $R$ containing $I$:

$$\sqrt{I} = \pi^{-1} \bigcap_{I \leq \mathfrak{p}} \overline{\mathfrak{p}} = \bigcap_{I \leq \mathfrak{p}} \pi^{-1}\overline{\mathfrak{p}} = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p}.$$ ■

**1.2.15 Definition**

Let $R$ be a ring, then define its **Jacobson radical** to be

$$J(R) = \bigcap_{\mathfrak{m}} \mathfrak{m},$$

the intersection of all maximal ideals.

**1.2.16 Lemma**

Let $\mathfrak{m} \leq R$ be maximal. Then $x \notin \mathfrak{m}$ iff there is a $y \in R$ such that $1 - xy \in \mathfrak{m}$.

**Proof**

$x \notin \mathfrak{m}$ iff $\overline{x} \in R/\mathfrak{m}$ is nonzero, or equivalently is a unit. This is iff there is a $y \in R$ such that $\overline{xy} = \overline{1}$. This just means $1 - xy \in \mathfrak{m}$, as desired. ■

**1.2.17 Corollary**

For a ring $R$, its Jacobson radical $J(R)$ is the set of all elements $x \in R$ such that $1 - yx \in R^\times$ for all $y \in R$.

**Proof**

Now if $x \notin J(R)$ then $x \notin \mathfrak{m}$ for some $\mathfrak{m}$, and then $1 - yx \in \mathfrak{m}$ for some $y$ and thus is not a unit. And if $1 - yx$ is not a unit, then $1 - yx \in \mathfrak{m}$ for some maximal $\mathfrak{m}$ and thus $x \notin \mathfrak{m}$ so that $x \notin J(R)$. ■

**1.2.18 Definition**

A ring $R$ is **local** if it has a unique maximal ideal $\mathfrak{m}$. The field $k = R/\mathfrak{m}$ is called its **residue field**. By abuse of notation we write $(R, \mathfrak{m})$ for a local field.

**1.2.19 Proposition**

**(1)** If $(R, \mathfrak{m})$ is a local ring then $R - \mathfrak{m} \subseteq R^\times$.

**(2)** Conversely if $I < R$ is a proper ideal with $R - I \subseteq R^\times$ then $(R, I)$ is a local ring.

**(3)** If $\mathfrak{m} \leq R$ is maximal such that $1 + \mathfrak{m} \subseteq R^\times$ then $(R, \mathfrak{m})$ is local.

**Proof**

If $x \notin \mathfrak{m}$ then $(x)$ is not contained in $\mathfrak{m}$, and since $\mathfrak{m}$ is the unique maximal ideal this means that $(x) = R$ so that $x$ is a unit. Also we noted that $R^\times$ is the intersection of all complements of maximal ideals, so this follows from that as well.

Conversely, we have that

$$R - I \subseteq R^\times \bigcap_{\mathfrak{m}} \mathfrak{m}^c \implies I \supseteq \bigcup_{\mathfrak{m}} \mathfrak{m},$$

which means that $I$ contains all maximal ideals, which can only happen if it is the unique maximal ideal.

We want to show that $R - \mathfrak{m} \subseteq R^\times$. Indeed, if $x \notin \mathfrak{m}$ then there is a $y$ such that $1 - yx \in \mathfrak{m}$ so that $yx \in 1 + \mathfrak{m}$ and is thus a unit. So $x$ is a unit, as desired. ■

**1.2.20 Example**

For $p \in \mathbb{Z}$ prime, define

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \,\middle|\, a, b \in \mathbb{Z}, b \notin (p) \right\} \subseteq \mathbb{Q}, \qquad I = \left\{ \frac{a}{b} \in \mathbb{Z}_{(p)} \,\middle|\, a \in (p) \right\}.$$

Then $(\mathbb{Z}_{(p)}, I)$ is local. Indeed, if $\frac{a}{b} \notin I$ then it is invertible (its inverse is $\frac{b}{a}$). We can generalize this, as we will in the next section.

## 1.3 Localization of rings

Consider the following problem: for a ring $A$ and a subset $S \subseteq A$, we want a ring $S^{-1}A$ and a morphism $\iota\colon A \to S^{-1}A$ such that any morphism $f\colon A \to B$ such that $f(S) \subseteq B^\times$, $f$ factors uniquely through $\iota$. That is, there is a unique $\hat{f}\colon S^{-1}A \to B$ such that $f = \hat{f} \circ \iota$:

$$\begin{tikzcd} A \arrow[r, "\iota"'] \arrow[dr, "f"'] & S^{-1}A \arrow[d, dashed, "\exists!"] \\ & B \end{tikzcd}$$

This is a universal property, and as such any solution is unique up to unique isomorphism. In this section we will investigate the construction of such a solution.

**1.3.1 Definition**

A subset $S \subseteq A$ is **multiplicatively closed** if $1 \in S$ and for $a, b \in S$ we have $ab \in S$.

It is sufficient to consider only the case that $S$ is multiplicatively closed. Otherwise, let $\hat{S}$ be $S$'s multiplicative closure: the set of all finite products of elements from $S$. Then $\hat{S}^{-1}R$ is a solution to the problem for $S$. Indeed, if $f(S) \subseteq B^\times$, then $f(\hat{S}) \subseteq B^\times$ as well, as the product of units. Then there is a unique factoring through $\hat{S}$'s $\iota$.

**1.3.2 Definition**

For $S \subseteq A$ multiplicatively closed, define an equivalence on $A \times S$ by $(a, s) \sim (b, t)$ iff there is a $u \in S$ such that $u(at - bs) = 0$, in which case we write $(a, s) \sim_u (b, t)$.

**1.3.3 Lemma**

This is indeed an equivalence relation.

**Proof**

The subrelations $\sim_u$ are all clearly reflexive and symmetric, thus their union $\sim$ is as well. Now if $(a, s) \sim_\alpha (b, t) \sim_\beta (c, v)$ then $\alpha at = \alpha sb$ and $\beta bv = \beta ct$. So then $\alpha t \beta av = \alpha sb\beta v = \alpha\beta tcs$ so that $(a, s) \sim_{\alpha t\beta} (c, v)$. ■

**1.3.4 Example**

A more naive equivalence would be $\sim_1$, i.e. $(a, s) \sim (b, t)$ iff $at - bs = 0$. This corresponds to $\sim$ when $A$ is an integral domain, but is not generally an equivalence.

**1.3.5 Definition**

For $S \subseteq A$ multiplicatively closed, define $S^{-1}A$ to be the quotient of $A \times S$ by $\sim$. Write the equivalence class of $(a, s)$ by $a/s$. We give this a ring structure by

$$0 = 0/1, \qquad 1 = 1/1, \qquad a/s + b/t = (at + bs)/st, \qquad a/s \cdot b/t = (ab)/(st).$$

Define $\iota\colon A \to S^{-1}A$ by $\iota(a) = a/1$.

**1.3.6 Proposition**

$S^{-1}A$ is indeed a ring, and $\iota$ is a ring morphism.

**Proof**

We need to show that addition and multiplication are well-defined. Indeed, if $(a, s) \sim_u (a', s')$ and $(b, t) \sim_v (b', t')$ then $(at + bs, st) \sim_{uv} (a't' + b's', s't')$ which can be verified. And similarly $(ab, st) \sim_{uv} (a'b', s't')$. From well-definedness it is easy to see that addition and multiplication define a ring structure and $\iota$ is a ring morphism. ■

A few properties are immediate:

**(1)** For $s \in S$, $\iota(s) = 1/s$ has an inverse $s/1$, so that $\iota(S) \subseteq (S^{-1}R)^\times$;

**(2)** We have that $a/s = 0/1$ iff there is a $t \in S$ such that $at = 0$. In particular, $\iota(a) = 0$ iff there is a $t \in S$ such that $ta = 0$.

**(3)** Every element of $S^{-1}A$ is of the form

$$a/s = a/1 \cdot 1/s = \iota(a) \cdot \iota(s)^{-1}.$$

**1.3.7 Theorem**

This construction solves the above universal property.

**Proof**

Let $f\colon A \to B$ be such that $f(S) \subseteq B^\times$: we need to show that there is a unique map $g\colon S^{-1}A \to B$ such that $f = g \circ \iota$. Indeed, we must define

$$g(a/s) = g(\iota(a)\iota(s)^{-1}) = (g \circ \iota(a)) \cdot (g \circ \iota(s))^{-1} = f(a)f(s)^{-1}.$$

This shows that $g$ is unique, if it exists. We need to show that this map is well-defined and a ring morphism. If $a/s = b/t$ then we must have that $f(a)f(s)^{-1} = f(b)f(t)^{-1}$ so that $g$ is well-defined. Indeed, if $u(at - bv) = 0$ for $u \in S$, we have

$$f(u) \cdot (f(at) - f(bv)) = 0$$

and since $f(S) \subseteq B^\times$ we have that $f(u)$ is a unit so that $f(at) - f(bv)$, meaning this is well-defined. And this is a ring morphism, as can be checked easily. ■

**1.3.8 Corollary**

Let $f\colon A \to B$ be a ring morphism such that

**(1)** $f(S) \subseteq B^\times$,

**(2)** if $f(a) = 0$ then $ta = 0$ for some $t \in S$,

**(3)** every element of $B$ is of the form $f(a)f(s)^{-1}$ for $a \in A$ and $s \in S$.

Then there is a unique isomorphism $g\colon S^{-1}A \to B$ such that $g \circ \iota = f$.

**Proof**

$f$ solves the universal problem above for $S$ as well. If $h\colon A \to C$ is a map where $h(S) \subseteq C^\times$, then we need to define $\hat{h}\colon B \to C$ by

$$\hat{h}(f(a)f(s)^{-1}) = h(a)h(s)^{-1},$$

which is well-defined and a ring morphism for similar reasons as above. Since universal properties are unique up to unique isomorphism, the result follows.

Alternatively we can prove this directly: let $g\colon S^{-1}A \to B$ be the unique map where $g \circ \iota = f$ by the universal property. By definition $g(a/s) = f(a)f(s^{-1})$. By the third point $g$ is surjective, and it is injective since if $g(a/s) = 0$ we have $f(a) = 0$ so that $ta = 0$, which means that $a/s = 0$. ■

**1.3.9 Example**

If $A$ is an integral domain, then $S = A - 0$ is multiplicatively closed, and $S^{-1}A$ is the field of fractions of $A$.

**1.3.10 Proposition**

For $A \neq 0$, then $S^{-1}A = 0$ iff $0 \in S$.

**Proof**

We have that $S^{-1}A = 0$ iff $1/1 = 0/1$ iff there is a $s \in S$ such that $s \cdot 1 = 0$. ■

**1.3.11 Definition**

Let $\mathfrak{p} \subseteq A$ be prime. Then $S = A - \mathfrak{p}$ is multiplicatively closed, and we denote its localization by $A_\mathfrak{p} = S^{-1}A$.

For $a \in A$, $S = \{1, a, a^2, \ldots\}$ is multiplicatively closed, and we denote its localization by $A_a = S^{-1}A$.

**Note:**

Note that we have two distinct localizations here, which have similar names in some cases. If $A$ is a PID and $p \in A$ is irreducible, then $A_{(p)}$ is localization by $A - (p)$, which $A_p$ is localization by $\{p^n\}_{n<\omega}$. Elements of the former are of the form $a/s$ where $p$ does not divide $s$, while elements of the former are of the form $a/s$ where $s = p^n$. Quite different!

**1.3.12 Definition**

For a ring morphism $f\colon A \to B$ and an ideal $I \subseteq A$, define the ideal $IB = (f(I))$ of $B$. Explicitly, $IB$ is the set of all finite sums $\sum_i f(a_i)b_i$ for $a_i \in I$.

In general we have $f^{-1}(f(I)) \supseteq I$ and $f(f^{-1}(J)) \subseteq J$. These imply $f^{-1}(IB) \supseteq I$ and $f^{-1}(J)B \subseteq J$.

**1.3.13 Definition**

For two ideals $I, J$, define $(I : J) = \{x \in A \mid xJ \subseteq I\}$. This is an ideal containing $I$. For $a \in A$, define $(I : a) = (I : (a)) = \{x \in A \mid xa \subseteq I\}$.

For $S \subseteq R$ multiplicatively closed, define the ideal $S^{-1}I = IS^{-1}A \subseteq S^{-1}A$. This is the ideal generated by $\iota(I)$.

So we have two maps: $S^{-1}\bullet$ mapping ideals of $A$ to ideals of $S^{-1}A$, and $\iota^{-1}$ mapping ideals of $S^{-1}A$ to $A$:

$$\{\text{Ideals of } A\} \underset{\iota^{-1}}{\overset{S^{-1}}{\rightleftarrows}} \left\{\text{Ideals of } S^{-1}A\right\}.$$

We wish to relate the two.

**1.3.14 Proposition**

**(1)** For an ideal $I \subseteq A$, $S^{-1}I = \{a/s \mid a \in I, s \in S\}$.

**(2)** For an ideal $J \subseteq S^{-1}A$, we have $S^{-1}\iota^{-1}J = J$.

**(3)** For an ideal $I \subseteq A$, we have $\iota^{-1}S^{-1}I = \bigcup_{s\in S}(I : s)$.

**(4)** We have $S^{-1}I = S^{-1}A$ iff $I \cap S \neq \emptyset$.

**(5)** For $\mathfrak{p} \subseteq A$ prime, if $I \cap S \neq \emptyset$ then $S^{-1}\mathfrak{p} \subseteq S^{-1}A$ is prime (otherwise it is all of $S^{-1}A$).

**(6)** The correspondence $\mathfrak{q} \mapsto \iota^{-1}\mathfrak{q}$ is a bijection between prime ideals of $S^{-1}A$ and prime ideals of $A$ not intersecting $S$. Its inverse is $S^{-1}$. That is, we have a one-to-one correspondence:

$$\{\text{Prime ideals of } A \text{ not intersecting } S\} \underset{\iota^{-1}}{\overset{S^{-1}}{\rightleftarrows}} \left\{\text{Prime ideals of } S^{-1}A\right\}.$$

**Proof**

**(1)** $\{a/s \mid a \in I, s \in S\}$ is an ideal and contains $\iota(I)$. It is clearly closed under multiplication by $S^{-1}A$ and for $a/s + b/t = (at + bs)/(st)$, notice that $at + bs \in I$. By definition $S^{-1}I$ is the smallest ideal containing $\iota(I)$, and it consists of finite sums $\sum_i \iota(a_i)x_i/s_i$. This clearly contains $a/s$, and thus they must be equal.

**(2)** As explained before, in general we have $S^{-1}\iota^{-1}J = \iota^{-1}(J)S^{-1}A \subseteq J$, we must prove the opposite inclusion. As above, we have $S^{-1}\iota^{-1}J = \{a/s \mid \iota(a) \in J, s \in S\}$. For $x/s \in J$, we have that $\iota(x) = x/1 \in J$ and so $x/s \in S^{-1}\iota^{-1}J$ as desired.

**(3)** If $a \in \iota^{-1}S^{-1}I$ then $\iota(a) \in S^{-1}I$ so by the first point $a/1 = b/s$ for $b \in I$. Then $asu = bu$ for some $u \in S$ and since $bu \in J$ this means that $a \in (I : su)$. Conversely if $a \in (I : s)$ for $s \in S$ then $as \in I$ and so $\iota(a) = a/1 = (as)/s \in S^{-1}I$.

**(4)** If $S^{-1}I = S^{-1}A$ then $1/s = a/t$ for $a \in I$, so $as = t$ meaning $as \in I \cap S$. Conversely if $a \in I \cap S$ then $1/1 = a/a \in S^{-1}I$.

**(5)** If $S \cap \mathfrak{p} = \emptyset$ then we need to show that $S^{-1}\mathfrak{p}$ is prime. If $a_1/s_1, a_2/s_2 \notin S^{-1}\mathfrak{p}$ we want to show $(a_1a_2)/(s_1s_2) \notin S^{-1}\mathfrak{p}$. By assumption $a_1, a_2 \notin \mathfrak{p}$.
Now we claim that if $a \notin \mathfrak{p}$, $a/s \notin S^{-1}\mathfrak{p}$, which would complete the proof. Assume the contrary: $a/s = b/t$ for $b \in \mathfrak{p}$ then there is a $u \in S$ such that $atu = bsu$. Then $atu \in \mathfrak{p}$, but $a \notin \mathfrak{p}$ and $t, u \in S$ and thus also are not in $\mathfrak{p}$, a contradiction to $\mathfrak{p}$'s primality.

**(6)** Preimages of prime ideals are prime, so $\iota^{-1}$ maps prime ideals to prime ideals. By the second point $S^{-1}\iota^{-1}\mathfrak{q} = \mathfrak{q}$ for all prime ideals $\mathfrak{q} \subseteq S^{-1}A$. It remains to show that $\iota^{-1}S^{-1}\mathfrak{p} = \mathfrak{p}$ for all prime $\mathfrak{p} \subseteq A$ which don't intersect $S$. We know that $\mathfrak{p} \subseteq \iota^{-1}S^{-1}\mathfrak{p}$ as is true in general, so we want to show that $\iota^{-1}S^{-1}\mathfrak{p} = \bigcup_{s\in S}(\mathfrak{p} : s)$ is contained in $\mathfrak{p}$. Indeed, if $a \in (\mathfrak{p} : s)$ then $as \in \mathfrak{p}$ and since $s \notin \mathfrak{p}$ we have that $a \in \mathfrak{p}$ as desired. ■

**1.3.15 Corollary**

Let $\mathfrak{p} \subseteq A$ be prime, and consider its localization $A_\mathfrak{p}$. Then $A_\mathfrak{p}$ is a local ring with maximal ideal $\mathfrak{p}A_\mathfrak{p}$. Moreover, the correspondence $\mathfrak{q} \to \mathfrak{q}A_\mathfrak{p}$ is a bijection between prime ideals of $A_\mathfrak{p}$ and prime ideals $\mathfrak{q} \subseteq \mathfrak{p}$.

**Proof**

Let $S = A - \mathfrak{p}$, so that $A_\mathfrak{p} = S^{-1}A$, and $\mathfrak{p}A_\mathfrak{p} = S^{-1}\mathfrak{p}$. The last point is clear: $\mathfrak{q} \cap S = \emptyset$ iff $\mathfrak{q} \subseteq \mathfrak{p}$. $A_\mathfrak{p}$ being local is also clear: $S^{-1}\bullet$ is order-preserving, so $S^{-1}\mathfrak{q} \subseteq S^{-1}\mathfrak{p}$ for all prime ideals $\mathfrak{q}$. Thus $S^{-1}\mathfrak{p}$ is maximal. ■

**1.3.16 Corollary**

If $a \in A$ is not nilpotent, then there is a prime ideal $\mathfrak{p} \subseteq A$ such that $a \notin \mathfrak{p}$.

**Proof**

As before, consider the multiplicative set $S = \{a^n\}_{n<\omega}$, and $A_a = S^{-1}A$. Since $a$ is not nilpotent, $0 \notin S$, so that $A_a \neq 0$. So take a prime ideal $\mathfrak{p} \subseteq A_a$, and then $\iota^{-1}\mathfrak{p}$ is a prime ideal of $A$ not intersecting $S$, thus not containing $a$. ■

**Note:**

This completes the proof before that

$$\operatorname{nil}(A) = \bigcap_{\mathfrak{p}} \mathfrak{p}$$

but without appealing to Zorn, and thus not relying on the Axiom of Choice.

## 1.4 Modules

**1.4.1 Definition**

Let $A$ be a ring, then a **module** over $A$ is an Abelian group $M$ along with a ring morphism $\phi\colon A \to \operatorname{end}(M)$. We define **multiplication** to be a function $\mu\colon A \times M \to M$, written by juxtaposition, by

$$\mu(a, m) = \phi(a)m.$$

An $A$-module can also be defined via multiplication: require that multiplication distributes both ways, and $1 \in A$ is an identity:

$$(a+b)m = am + bm, \qquad a(m+n) = am + an, \qquad a(bm) = (ab)m, \qquad 1m = m.$$

**1.4.2 Definition**

For two $A$-modules $M$ and $N$, an **$A$-module morphism** is a map $f\colon M \to N$ which is an Abelian group morphism and also multiplicative:

$$f(am) = af(m)$$

for all $a \in A, m \in M$.

**Note:**
I'm going to skip the rest of the standard definitions, for Chrissake.

**1.4.3 Definition**
If $M$ is an $A$-module, we define $\mathrm{ann}_A(M) \leq A$ the **annihilator** of $M$ by

$$\mathrm{ann}_A(M) = \{a \in A \mid aM = 0\} = \ker\phi,$$

where $\phi$ is the module map $\phi\colon A \to \mathrm{end}(M)$. This is then an ideal. For $S \subseteq M$ a subset we define $\mathrm{ann}_A(S)$ similarly.
A module $M$ is **faithful** if $\mathrm{ann}_A(M) = 0$.

$\mathrm{ann}_A(S)$ is equal to $\mathrm{ann}_A(\langle S \rangle)$, the annihilator of the submodule generated by $S$ (all $R$-combinations of elements from $S$), and is thus also an ideal.

**1.4.4 Definition**
For a ring $A$, an **$A$-algebra** is a ring $B$ equipped with a ring morphism $\phi\colon A \to B$.

Note that $B$ can be embedded into $\mathrm{end}(B)$ mapping $b \in B$ to multiplication by $b$. Thus an $A$-algebra is an $A$-module with multiplication given by $a \cdot b = \phi(a)b$.

**1.4.5 Definition**
For two $A$-algebras $B, C$, an **$A$-algebra morphism** is a map $f\colon B \to C$ which is both a ring morphism and an $A$-module morphism.

If $\phi\colon A \to B, \psi\colon A \to C$ are the algebra maps, then $f\colon B \to C$ is an algebra morphism iff it is a ring morphism and $\psi = f \circ \phi$:

$$\psi(a) = a1_C = af(1_B) = f(a1_B) = f \circ \phi(a).$$

**1.4.6 Definition**
For an $A$-module $M$ and a submodule $N$, its quotient module $M/N$ is the set of cosets $m + N$ with the usual $M$-module structure. The usual isomorphism results hold.

**1.4.7 Proposition**
Let $I \leq A$ be a module, then every $A/I$-module is an $A$-module with $\mathrm{ann}_A(M) \supseteq I$ and vice-versa. In particular an $A$-module $M$ can be viewed as a faithful $A/\mathrm{ann}_A(M)$-module.

**Proof**

Consider the projection $\pi\colon A \to A/I$. Composing with $\phi\colon A/I \to \mathrm{end}(M)$ gives an $A$-module structure to an $A/I$-module and $\ker(\phi \circ \pi) \supseteq \ker\phi = I$. Conversely if $\phi\colon A \to \mathrm{end}(M)$ has kernel containing $I$ we can quotient it by $I$. ■

**1.4.8 Definition**

For a collection of $A$-modules $\{M_\alpha\}_\alpha$ we define their **direct product** and **direct sum** by

$$\prod_\alpha M_\alpha = \{(m_\alpha) \mid m_\alpha \in M_\alpha\}, \qquad \bigoplus_\alpha M_\alpha = \left\{ (m_\alpha) \in \prod_\alpha M_\alpha \;\middle|\; \text{All but finitely many of } m_\alpha \text{ are zero} \right\}.$$

$\prod_\alpha M_\alpha$'s module structure is given by pointwise operations, and $\bigoplus_\alpha M_\alpha$ is viewed as a submodule. These form the products and coproducts in the category of $A$-modules.

**1.4.9 Definition**

An $A$-module $M$ is **free** if

$$M \cong A^{\oplus S} = \bigoplus_{\alpha \in S} A$$

for some set $S$. Note that $\{e_\alpha\}_{\alpha \in S}$ (the usual standard basis vectors) forms a basis of $A^{\oplus S}$ in the sense that every element of $A^{\oplus S}$ can be written as a unique combination of standard basis vectors. A collection of elements $\{m_\alpha\}_{\alpha \in S}$ is a **basis** of $M$ if mapping $m_\alpha \mapsto e_\alpha$ extends to an isomorphism $M \cong A^{\oplus S}$.

A basis of $M$ is equivalently a set of elements which generate $M$ and are also linearly independent in the usual way.

A free module is called such because they are free in the usual sense with respect to the obvious forgetful functor $\mathrm{Mod}_A \to \mathrm{Set}$. That is, $\hom_A(A^{\oplus S}, B) \cong \hom_{\mathrm{Set}}(S, B)$.

**1.4.10 Proposition**

If $A^{\oplus S} \cong A^{\oplus S'}$ then $S$ and $S'$ have the same cardinality.

**Proof**

Let $\mathfrak{m} \leq A$ be maximal, then the isomorphism $A^{\oplus S} \cong A^{\oplus S'}$ induces an isomorphism $(A/\mathfrak{m})^{\oplus S} \cong (A/\mathfrak{m})^{\oplus S'}$. $A/\mathfrak{m}$ is a field, and so this is an isomorphism of $A/\mathfrak{m}$-modules (vector spaces) of dimensions $|S|, |S'|$. Since the dimension is invariant, their cardinalities must be equal. ■

**1.4.11 Definition**

If $M$ is a free $A$-module, then its **rank** is the cardinality of $S$ where $M \cong A^{\oplus S}$.

**1.4.12 Definition**

An $A$-module $M$ is **finitely generated** (**fg**) if there is a finite set $S \subseteq M$ such that $M = \langle S \rangle$.

**1.4.13 Proposition**

An $A$-module is finitely generated iff it is a quotient of $A^n$ for some $n$ (up to isomorphism).

**Proof**

If $\phi\colon A^n \to M$ is surjective, then $\phi(e_i)$ generate $M$ clearly and so it is finitely-generated. Thus if $M \cong A^n/N$ is a quotient of $A^n$ then the projection $\pi\colon A^n \to A^n/N \cong M$ is surjective and thus $M$ is finitely-generated. Conversely let $m_i$ generate $M$, and define $\phi\colon A^n \to M$ by $\phi(e_i) = m_i$ which extends by linearity to all of $A^n$ since it is free. This is surjective and then $A^n/\ker\phi \cong M$ as desired. ■

**1.4.14 Proposition**

Let $M$ be a fg $A$-module, and $I \leq A$ an ideal such that $M = IM$. Then there is an $x \in I$ such that $(1-x)M = 0$ ($xm = m$ for all $m \in M$).

**Proof**

Let $m_1, \ldots, m_n$ generate $M$, we induct on $n$. For $n = 0$ we have $M = 0$ so this is clear. Consider $N = M/(m_n)$, then $IN = (IM + (m_n))/(m_n) = M/(m_n) = N$ and $N$ is generated by $n-1$ elements, so by induction there is an $x \in I$ such that $(1-x)N = 0$. Meaning $(1-x)M \subseteq (m_n)$, and since $m_n \in M = IM$ we have

$$(1-x)m_n \in (1-x)IM = I(1-x)M \subseteq I(m_n),$$

so $(1-x)m_n = ym_n$ for some $y \in I$ and thus $(1-x-y)m_n = 0$. Now $(1-x-y)(1-x) \in 1+I$ and satisfies

$$(1-x-y)(1-x)M \subseteq (1-x-y)(m_n) = 0$$

as desired. ■

**1.4.15 Corollary (Nakayama's lemma)**

Let $M$ be a fg $A$-module, such that $J(A)M = M$, then $M = 0$.

**Proof**

By above there is an $x \in J(A)$ such that $(1-x)M = 0$. Since $x \in J(A)$, $1-x \in A^\times$ so that $(1-x)M = M = 0$ as desired. ■

**1.4.16 Corollary**

Let $M$ be a fg $A$-module and $N \subseteq M$ a submodule such that $M = IM + N$ for some $I \subseteq J(A)$, then $M = N$.

**Proof**

Consider $M/N$, then $I(M/N) = (IM+N)/N = M/N$, so by above $M/N = 0$ meaning $M = N$. ■

**1.4.17 Corollary**

Let $(A, \mathfrak{m})$ be local, $M$ a fg $A$-module. Then $x_1, \ldots, x_n \in M$ generate $M$ iff $\overline{x}_1, \ldots, \overline{x}_n \in M/\mathfrak{m}M$ generate $M/\mathfrak{m}M$ over $A/\mathfrak{m}$.

**Proof**

If $x_i$ generate $M$ then $\overline{x}_i$ clearly generate $M$. Conversely, let $N = (x_1, \ldots, x_n)$ be the submodule generated by $M$, and consider the composition $f\colon N \to M \to M/\mathfrak{m}M$. $f$'s image is the span of $(\overline{x}_i)_i$, so all of $M/\mathfrak{m}$. Thus $f$ is surjective, so $M = N + \mathfrak{m}M$, since for every $m \in M$ there is an $n \in N$ such that $f(n) - m \in \mathfrak{m}M$ by surjectivity, and by above since $J(A) = \mathfrak{m}$ this means $M = N$ as desired. ■

**1.4.18 Definition**

Recall the definition of an exact sequence:

$$M \xrightarrow{f} N \xrightarrow{g} P$$

is **exact** at $N$ if $\mathrm{im} f = \ker g$.

## 1.5 Localization of modules

**1.5.1 Definition**

Let $S \subseteq A$ be multiplicatively closed and $M$ an $A$-module. Define an equivalence on $M \times S$ by $(m, s) \sim (m', s')$ iff there is a $t \in S$ such that $t(s'm - sm') = 0$. This is an equivalence relation and we denote the equivalence class of $(m, s)$ by $m/s$. The quotient $S^{-1}M$ forms an $A$-module by

$$m/s + m'/s' = (s'm + sm')/(ss'), \qquad a \cdot (m/s) = (am)/s.$$

Note that localization at $S$ forms an additive functor $S^{-1}\colon \mathrm{Mod}_A \to \mathrm{Mod}_{S^{-1}A}$. Indeed, $S^{-1}M$ is an $S^{-1}A$ module by

$$a/s \cdot m/s' = (am)/(ss').$$

And if $f\colon M \to N$ is an $A$-module morphism, then define

$$S^{-1}f\colon S^{-1}M \to S^{-1}N, \qquad S^{-1}f(m/s) = f(m)/s.$$

It is not hard to see that this is functorial and additive.

There is the canonical morphism $\iota\colon A \to S^{-1}A$ and as such it lifts to a "forgetful" functor $\mathrm{Mod}_{S^{-1}A} \to \mathrm{Mod}_A$, where for $M \in \mathrm{Mod}_{S^{-1}A}$ we view it as an $A$-module by $a \cdot M = \iota(a) \cdot M$.

**1.5.2 Proposition**

The functor $S^{-1}\colon \mathrm{Mod}_A \to \mathrm{Mod}_{S^{-1}A}$ is left adjoint to the forgetful functor $\mathrm{Mod}_{S^{-1}A} \to \mathrm{Mod}_A$.

**Proof**

We need to show that there is a canonical bijection

$$\hom_{S^{-1}A}(S^{-1}M, N) \cong \hom_A(M, N),$$

for all $A$-modules $M$ and $S^{-1}A$-modules $N$. For $f\colon M \to N$ map it to $\hat{f}\colon S^{-1}M \to N$ by $\hat{f}(m/s) = 1/s \cdot f(m)$. And for $g\colon S^{-1}M \to N$ map it to $\tilde{g}\colon M \to N$ by $\tilde{g}(m) = g(m/1)$. ■

**1.5.3 Definition**

An additive functor $F\colon \mathrm{Mod}_A \to \mathrm{Mod}_B$ is

**(1)** **left exact** if it preserves left-exact sequences: if

$$0 \to M \xrightarrow{f} N \xrightarrow{g} P$$

is exact, then so is

$$0 \to FM \xrightarrow{Ff} FN \xrightarrow{Fg} FP.$$

**(2)** **right exact** if it preserves right-exact sequences: if

$$M \xrightarrow{f} N \xrightarrow{g} P \to 0$$

is exact, then so is

$$FM \xrightarrow{Ff} FN \xrightarrow{Fg} FP \to 0.$$

**(3)** **exact** if it is both left and right exact.

**1.5.4 Proposition**

An additive functor is

**(1)** left exact iff it preserves kernels iff it preserves finite limits iff it preserves the left side of a short exact sequence.

**(2)** right exact iff it preserves cokernels iff it preserves finite colimits iff it preserves the right side of a short exact sequence.

**(3)** exact iff it preserves finite limits and colimits iff it preserves short exact sequences iff it preserves all exact sequences.

**Proof**

Left-exact sequences are kernel diagrams, and right-exact sequences are cokernel diagrams. So by definition, left exact functors are those which preserve kernels, and right exact functors are those which preserve cokernels. Since additive functors preserve finite biproducts, left/right exact functors preserve limits/colimits.

Clearly left-exact functors preserve the left side of a short exact sequence (similarly for right-exact functors). To show the converse, if left-exact sequences can be quotiented to give an appropriate short exact sequence, similar for right-exact sequences. ■

**1.5.5 Corollary**

Left adjoints are right exact, and right adjoints are left exact.

**Proof**

Left adjoints preserve all colimits, and right adjoints all limits. ■

**1.5.6 Corollary**

$S^{-1}$ is an exact functor.

**Proof**

By the above corollary, it is right-exact. To show that it is exact, it is sufficient to show that it preserves injections. If $f\colon M \to N$ is injective then $m/s \in \ker S^{-1}f$ iff $f(m)/s = 0/1$, so that $tf(m) = 0$ for some $t \in S$. But then $tf(m) = f(tm)$ so $tm \in \ker\phi$, so $tm = 0$, which means $m/s = 0/1$ as desired. ■

In general if $U\colon \mathrm{Mod}_A \to \mathrm{Mod}_B$ is a restriction functor corresponding to a morphism $\iota\colon B \to A$, then it has a left adjoint via the tensor product. In particular, viewing $A$ as a $(A, B)$-module, then $A \otimes_B M$ has an $A$-module structure (given by $a(a' \otimes m) = (aa') \otimes m$). $U$'s left adjoint is then $A \otimes_B M$.

Since left adjoints are unique up to natural isomorphism,

$$S^{-1}M \cong S^{-1}A \otimes_A M.$$

Thus $S^{-1}A \otimes_A \bullet$ is an exact functor, which means that $S^{-1}A$ is a **flat** $A$-module. This can also be shown directly: $S^{-1}A$ is the filtered colimit of free $A$-modules, and is thus flat.

**1.5.7 Corollary**

$S^{-1}$ preserves arbitrary coproducts:

$$S^{-1}\bigoplus_\alpha M_\alpha \cong \bigoplus_\alpha S^{-1}M_\alpha$$

naturally.

**Proof**

Left adjoints preserve arbitrary colimits. ■

**Note:**
While $S^{-1}$ is exact, it is not **continuous**: it does not preserve arbitrary limits (equivalently it does not preserve arbitrary products). That is, we do *not* have that

$$S^{-1}\prod_\alpha M_\alpha \cong \prod_\alpha S^{-1}M_\alpha.$$

This is of course true when the index set is finite, since then it is just a biproduct.

**1.5.8 Corollary**

**(1)** If $N \subseteq M$ is a submodule, $S^{-1}N$ can be naturally embedded in $S^{-1}M$.

**(2)** There is a natural isomorphism $S^{-1}(M/N) \cong (S^{-1}M)/(S^{-1}N)$.

**(3)** For $f\colon M \to N$ of $A$-modules, we have natural isomorphisms

$$\ker S^{-1}f \cong S^{-1}\ker f, \qquad \operatorname{im} S^{-1}f \cong S^{-1}\operatorname{im} f, \qquad S^{-1}\operatorname{coker} f \cong \operatorname{coker} S^{-1}f.$$

**(4)** For every two submodules $M_1, M_2 \subseteq M$ under the natural embeddings,

$$S^{-1}(M_1 \cap M_2) = S^{-1}M_1 \cap S^{-1}M_2 \subseteq S^{-1}M.$$

**Proof**

The first point is due to $S^{-1}$ being exact and thus preserving injections. The second point is because $S^{-1}$ preserves the short exact sequence $0 \to N \to M \to M/N \to 0$. The third point is clear as well, with only im not being immediately obvious. $\operatorname{im} f$ can be characterized as $\operatorname{im} f = \ker\operatorname{coker} f$, so it follows.

For the last point, note that $M_1 \cap M_2$ is the kernel of $M_1 \oplus M_2 \to M$ by $(x, y) \mapsto x - y$ (via the diagonal embedding), and so this follows from $S^{-1}$'s exactness. ■

**1.5.9 Definition**

For a prime ideal $\mathfrak{p} \leq A$ and an $A$-module $M$, denote by $M_{\mathfrak{p}}$ the localization of $M$ by $S = A - \mathfrak{p}$.

**1.5.10 Lemma**

For an $A$-module $M$, consider the natural map

$$\phi\colon M \to \prod_{\mathfrak{m}} M_{\mathfrak{m}}, \qquad m \mapsto (m/1)_{\mathfrak{m}},$$

where the product is over all maximal ideals of $A$. Then $\phi$ is injective.

**Proof**

Assume there is an $0 \neq x \in M$ such that $\phi(x) = 0$. Then define

$$I = \operatorname{ann}_A(x) = \{a \in A \mid ax = 0\}.$$

$I$ is a proper ideal of $A$, and thus is contained in a maximal ideal $\mathfrak{m}$. Then for every $s \in A - \mathfrak{m}$ we have that $s \notin I$ and so $sx \neq 0$, and thus $m/1$ is not zero in $M_{\mathfrak{m}}$, a contradiction. ■

**1.5.11 Corollary**

For an $A$-module $M$, the following are equivalent:

**(1)** $M = 0$,

**(2)** $S^{-1}M = 0$ for every multiplicative set $S \subseteq A$,

**(3)** $M_{\mathfrak{p}} = 0$ for every prime ideal $\mathfrak{p} \leq A$,

**(4)** $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m} \leq A$.

**Proof**

Clearly each point implies the next. By above $M$ embeds into $\prod_{\mathfrak{m}} M_{\mathfrak{m}}$, so if $M_{\mathfrak{m}} = 0$ for all $\mathfrak{m}$ then $M = 0$; the last point implies the first. ■

**1.5.12 Corollary**

For a morphism of $A$-modules $f\colon M \to N$, the following are equivalent:

**(1)** $f$ is injective/surjective/an isomorphism,

**(2)** $S^{-1}f$ is injective/surjective/an isomorphism for every multiplicative $S \subseteq A$,

**(3)** $f_{\mathfrak{p}}$ is injective/surjective/an isomorphism for every prime $\mathfrak{p} \leq A$,

**(4)** $f_{\mathfrak{m}}$ is injective/surjective/an isomorphism for every maximal $\mathfrak{m} \leq A$.

**Proof**

By $S^{-1}$'s exactness, the first point implies the second, and clearly all the other points imply the next. To show the last implies the first, let $K = \ker f$ so that $0 \to K \to M \xrightarrow{f} N$ is exact. Then $0 \to K_{\mathfrak{m}} \to M_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}}} N_{\mathfrak{m}}$ is exact, and so $K_{\mathfrak{m}} = \ker f_{\mathfrak{m}} = 0$ for all $\mathfrak{m}$. By above this means $K = 0$ so that $f$ is injective. Similar for surjectivity. ■

## 1.6 Finitely generated, integral, and finite algebras

**1.6.1 Definition**

For an $A$-algebra $B$, and $S \subseteq B$ denote by $A[S]$ the smallest $A$-subalgebra containing $S$ in $B$.

Let $\phi\colon A \to B$ be the algebra map, then $A[S]$ is the subring of $B$ generated by $\phi(A) \cup S$. Alternatively, for every $s \in S$ define an indeterminant $x_s$, and let $X = \{x_s\}_{s \in S}$ be the set of all these indeterminants. Consider the polynomial algebra $A[X]$, and define $f\colon A[X] \to B$ by $f(x_s) = s$. Then $A[S]$ is just the image of $f$. Explicitly, for a function $\alpha\colon S \to \mathbb{Z}_{\geq 0}$ with finite support, define $S^{\alpha} = \prod_{\alpha(s) \neq 0} s^{\alpha(s)}$. Then

$$A[S] = \left\{ \text{Finite sums } \sum_{\alpha} \phi(a_\alpha) S^{\alpha} \;\middle|\; a_\alpha \in A \right\}.$$

Note that we can consider $B$ as an $A$-module. Then for $S \subseteq B$, we can consider the *submodule* generated by $S$: $\langle S \rangle$. Since $A[S]$ is itself a module, $\langle S \rangle \subseteq A[S]$ automatically, and we do not have equality in general.

**1.6.2 Definition**

An $A$-algebra $B$ is **finitely generated** if there is a finite $S \subseteq B$ such that $B = A[S]$. $B$ is a **finite algebra** if it is finitely generated as an $A$-module.

A finite $A$-algebra is also finitely generated as an $A$-algebra, but not vice versa in general.

**1.6.3 Example**

Suppose $S = \{s\}$ is a singleton, then $A[s]$ is the set of polynomials $\sum_k \phi(a_k)s^k$.

**1.6.4 Example**

Clearly the polynomial algebra $A[x_1, \dots, x_n]$ is a finitely generated $A$-algebra. But it is not a finite $A$-algebra: we can reduce this to the case of fields, in which $x_i^k$ are all linearly independent and thus there is no finite basis.

**1.6.5 Lemma**

An $A$-algebra $B$ is fg iff it is the quotient of some finite polynomial algebra $A[x_1, \dots, x_n]$ (up to isomorphism of $A$-algebras).

**Proof**

This is equivalent to saying $B$ is fg iff there is a surjection $A[x_1, \dots, x_n] \to B$. This is clearly true: if $B$ is fg then for $s_1, \dots, s_n$ generating $B$ map $x_i \mapsto s_i$ and extend by freeness. Conversely, quotients of fg algebras are fg. ■

**1.6.6 Definition**

Let $B$ be an $A$-algebra. An element $b \in B$ is **integral** over $A$ if $b$ is the root of a monic polynomial over $A$. That is, there exist $a_1, \dots, a_n \in A$ such that $b^n + a_1 b^{n_1} + \cdots + a_n \cdot 1 = 0$.

$b \in B$ is **algebraic** over $A$ if $b$ is the root of any polynomial over $A$.

Clearly integral elements are algebraic. For fields the converse holds as well, as we can divide a polynomial by its leading coefficient.

**1.6.7 Example**

The set of integral elements in the $\mathbb{Z}$-algebra $\mathbb{Q}$ is $\mathbb{Z}$. Clearly an integer is integral, as $n \in \mathbb{Z}$ is the root of the monic $x - n$. Conversely if $b = p/q \in \mathbb{Q}$ is integral with $p, q$ coprime then

$$b^n + a_1 b^{n-1} + \cdots + a_n 1 = 0, \qquad a_i \in \mathbb{Z}.$$

Then we have that

$$p^n + a_1 p^{n-1} q + \cdots + a_n q^n = 0$$

and so $q$ must divide $p^n$, but since $p, q$ are coprime this can only happen if $q = 1$ and $b \in \mathbb{Z}$.

**1.6.8 Lemma**

If $M$ is a fg $A$-module, then for every $\phi \in \mathrm{end}_A(M)$ there exist $a_1, \ldots, a_n \in A$ such that

$$\phi^n + a_1\phi^{n-1} + \cdots + a_n = 0$$

in $\mathrm{end}_A(M)$.

**Proof**

The proof is by a generalization of Cayley-Hamilton. Let $n \geq 1$ and consider $R = \mathbb{Z}[X_{ij}]$, the polynomial ring over $n^2$ indeterminants $X_{ij}$, $1 \leq i, j \leq n$. Let $X = (X_{ij}) \in M_n(R)$, and define its **characteristic polynomial**: $\Phi_X(T) \in R[T]$ given by

$$\Phi_X(T) = \det(T\mathrm{id} - X).$$

By Cayley-Hamilton over $R$'s field of fractions, $\Phi_X(X) = 0$.

Now if $A$ is any ring and $Y \in M_n(A)$, there is a unique ring morphism $\phi\colon R \to A$ which maps $X$ to $Y$ (coordinate-wise). Indeed, such a map must map $X_{ij}$ to $Y_{ij}$ which uniquely determines $\phi$. Then $\phi$ lifts to the ring morphism $R[T] \to A[T]$ (map $\sum_n a_n T^n$ to $\sum_n \phi(a_n)T^n$), which maps $\Phi_X(T)$ to $\Phi_Y(T)$ (which is also given by the determinant $\det(T\mathrm{id} - Y)$ since the determinant is functorial: $\det(\phi(B)) = \phi(\det(B))$). Thus $\Phi_X(X) = 0$ implies that $\Phi_Y(Y) = \phi(\Phi_X(\phi(X))) = \phi(\Phi_X(X)) = 0$, as desired. So Cayley-Hamilton applies to any commutative ring.

Now for our lemma, let $m_1, \ldots, m_n \in M$ be generators. Then for all $j$ we have

$$\phi(m_j) = \sum_i a_{ij} m_i, \qquad a_{ij} \in A.$$

We associate to $\phi$ the representative matrix $[\phi] = (a_{ij}) \in M_n(A)$ (which is not unique, unlike over fields). This has the usual properties that $[\phi^n] = [\phi]^n$ and $[\phi + \psi] = [\phi] + [\psi]$ (meaning that the composition or sum of representatives is a representative matrix). Now Cayley-Hamilton gives us that

$$\sum_n b_n [\phi]^n = 0$$

for some $b_n \in A$, and thus

$$\left[\sum_n b_n \phi^n\right] = 0,$$

and since only the zero endomorphism has the zero representative, this means $\sum_n b_n \phi^n = 0$ as desired. ■

**1.6.9 Proposition**

For an $A$-algebra $B$ and $b \in B$, the following are equivalent:

**(1)** $b$ is integral over $A$,

**(2)** $A[b] \subseteq B$ is a finite algebra (fg module) over $A$,

**(3)** $A[b]$ is contained in a finite algebra over $A$,

**(4)** There exists a faithful $A[b]$-module $M$ such that $M$ is finitely generated as an $A$-module.

**Proof**

Clearly (2) implies (3) implies (4): take $M = C$, and since $A[b] \subseteq M$ it is faithful (the algebra map is inclusion, which is injective). (1) implies (2) since if $b^n + a_1 b^{n-1} + \cdots + a_n 1 = 0$ then $b^n$ is in $\langle 1, \ldots, b^{n-1} \rangle \subseteq B$ and then by induction so is every higher power of $b$, so that $A[b] = \langle 1, \ldots, b^{n-1} \rangle$ is finite.

(4) implies (1): we have a ring morphism $\phi\colon A[b] \to \mathrm{end}_A(M)$, so that every $f \in A[b]$ gives us an endomorphism $\phi(f) \in \mathrm{end}_A(M)$. In particular by the above lemma there are $a_1, \ldots, a_n \in A$ such that

$$\rho(b)^n + a_1 \rho(b)^{n-1} + \cdots + a_n = 0 \in \mathrm{end}_A(M).$$

For $f(b) = b^n + a_1 b^{n-1} + \cdots + a_n \in A[b]$, we see that $\rho(f(b)) = 0$. But $M$ is a faithful $A[b]$-module, so $f(b) = 0$ and $b$ is integral as desired. ■

**1.6.10 Lemma**

**(1)** If $B$ is a finite $A$-algebra, and $M$ is a fg $B$-module, then $M$ is a fg $A$-module.

**(2)** If $B$ is a finite $A$-algebra, and $C$ is a finite $B$-algebra, then $C$ is a finite $A$-algebra.

**Proof**

Clearly the second point follows from the first. Now if $M$ is generated by $m_1, \ldots, m_n$ over $B$, and $B$ is generated by $b_1, \ldots, b_k$ over $A$ then $b_i m_j$ generates $M$ over $A$. Indeed, for $m \in M$ if $m = \sum_i b_i' m_i$, then $b_i' = \sum_j a_{ij} b_j$ so

$$m = \sum_{i,j} a_{ij}(b_j m_i),$$

as desired. ■

**1.6.11 Corollary**

Let $B$ be an $A$-algebra, and suppose that $b_1, \ldots, b_n \in B$ are integral over $A$. Then $A[b_1, \ldots, b_n]$ is a finite $A$-algebra.

**Proof**

By induction: for $n = 1$ this is our above proposition. Then if $b_{n+1}$ is integral over $A$, it is integral over $A[b_1, \ldots, b_n]$ so that $A[b_1, \ldots, b_{n+1}]$ is a finite $A[b_1, \ldots, b_n]$-algebra, and thus a finite $A$-algebra by transitivity which we just showed. ■

**1.6.12 Corollary**

Let $B$ be an $A$-algebra, then

$$B_0 = \{b \in B \mid b \text{ is integral over } A\}$$

is an $A$-subalgebra of $B$

**Proof**

Clearly $0, 1 \in B_0$. Now if $b_1, b_2 \in B_0$ then $A[b_1 + b_2], A[b_1 b_2] \subseteq A[b_1, b_2]$ which is a finite $A$-algebra and thus $b_1 + b_2, b_1 b_2$ are integral over $A$; $B_0$ is a subring of $B$. And if $a \in A, b \in B_0$ then $A[ab] \subseteq A[b]$ is finite, so

that $ab \in B_0$. ■

**1.6.13 Definition**

**(1)** For an $A$-algebra $B$, we define $A$'s **integral closure** in $B$ to be the set of all integral elements over $A$ in $B$; which we showed above is an $A$-subalgebra.

**(2)** An $A$-algebra $B$ is **integral** if every element of $B$ is integral over $A$.

**(3)** A morphism of rings $f\colon A \to B$ is **integral** if $B$ is an integral $A$-algebra via $f$.

**(4)** For a morphism of rings $f\colon A \to B$, $f(A)$ is **integrally closed** in $B$ if $B_0 = f(A)$ (i.e. no $b \in B - f(A)$ is integral over $A$).

**(5)** A domain $A$ is **normal** if it is integrally closed in its field of fractions.

**1.6.14 Example**

We saw that $\mathbb{Z}$ is normal (its field of fractions is $\mathbb{Q}$); the same proof shows that any UFD is normal. In particular $k[x_1, \ldots, x_n]$ and $\mathbb{Z}[x_1, \ldots, x_n]$ are normal.

Consider the $k$-subalgebra $A = k[x^2, x^3] \subseteq k[x]$. Then $x \notin A$ is integral over $A$ since it is a root of $t^6 - x^6 \in A[t]$, and $x = x^3/x^2$ is in its field of fractions. Thus $A$ is not normal.

**Note:**

There is similarity between integral morphisms and algebraic extensions of fields. But while algebraically closed fields exist, there are no "integrally closed ring": every ring can be embedded non-integrally into another ring.

**1.6.15 Corollary**

An $A$-alegbra $B$ is finite over $A$ iff it is integral and finitely generated.

**Proof**

If $B$ if a finite $A$-algebra, then for every $b \in B$, $A[b] \subseteq B$ is finite so $b$ is integral over $A$; $B$ is integral and finitely generated. Conversely if $B$ is integral and finitely generated, suppose $B = A[b_1, \ldots, b_n]$; since each $b_i$ is integral we have that $B$ is finite. ■

**1.6.16 Corollary**

Let $B$ be an integral $A$-algebra, and $C$ be a $B$-algebra.

**(1)** If $c \in C$ is integral over $B$, then it is integral over $A$.

**(2)** If $C$ is integral over $B$, then it is integral over $A$.

**Proof**

If $c$ is integral over $B$, then
$$c^n + b_1 c^{n-1} + \cdots + b_n = 0$$
for some $b_i \in B$, which are integral over $A$. Then $B' = A[b_1, \ldots, b_n] \subseteq B$ is a finite $A$-algebra, and $c$ is integral over $B'$ and so $B'[c]$ is a finite $B'$-algebra. Since $B'$ is a finite $A$-algebra, by transitivity $B'[c]$ is a finite $A$-algebra as well containing $A[c]$, so $c$ is integral over $A$.
The second point follows from the first. ■

**1.6.17 Corollary**

Let $B$ be an $A$-algebra, and $B_0$ be the integral closure of $A$ inside $B$. Then $B_0$ is integrally closed in $B$.

**Proof**

If $b \in B$ is integral over $B_0$, then by the above corollary, since $B_0$ is an integral $A$-algebra, $b$ is integral over $A$ and thus $b \in B_0$. ■

Let $B$ be an $A$-algebra under $f\colon A \to B$.

**(1)** For an ideal $J \leq B$, define $J_A = f^{-1}(J) \leq A$. Then $f$ induces an injective homomorphism $A/J_A \to B/J$, so that $B/J$ is an $A/J_A$-algebra.

**(2)** For a multiplicative set $S \subseteq A$, $f(S) \subseteq B$ is multiplicative. Then we have a natural morphism $S^{-1}A \to f(S)^{-1}B$ mapping $a/s$ to $f(a)/f(s)$. Thus $f(S)^{-1}B$ is an $S^{-1}A$-algebra.

**(3)** Viewing $B$ as an $A$-module, we have already defined $S^{-1}B$, and this clearly coincides with $f(S)^{-1}B$. Thus $S^{-1}B$ is naturally an $S^{-1}A$-algebra.

**1.6.18 Proposition**

Let $B$ be an integral $A$-algebra.

**(1)** Let $J \leq B$ be an ideal, then $B/J$ is an integral $A/J_A$-algebra.

**(2)** Let $S \subseteq B$ be multiplicative, then $S^{-1}B$ is an integral $S^{-1}A$-algebra.

**Proof**

For $\overline{x} \in B/J$, lift it to $x \in B$, which is integral over $A$. Thus $x^n + a_1 x^{n-1} + \cdots + a_n = 0$ for $a_i \in A$, and then
$$\overline{x}^n + \overline{a}_1 \overline{x}^{n-1} + \cdots + \overline{a}_n = 0,$$
where $\overline{a}_i = a_i + J_A \in A/J_A$, so that $\overline{x}$ is integral over $A/J_A$.
An element of $S^{-1}B$ is of the form $\tilde{x} = x/f(s)$ with $x \in B, s \in S$. By assumption, $x^n + a_1 x^{n-1} + \cdots + a_n = 0$ and thus
$$\left(\frac{x}{f(s)}\right)^n + \frac{a_1}{s}\left(\frac{x}{f(s)}\right)^{n-1} + \cdots + \frac{a_n}{s^n} = \frac{x^n + a_1 x^{n-1} + \cdots + a_0}{f(s)^n} = 0,$$
so that $\tilde{x}$ is integral over $S^{-1}A$ as desired. ■

**1.6.19 Proposition**

Let $A \subseteq B$ be integral domains so that $B$ is integral over $A$. Then $A$ is a field iff $B$ is a field.

**Proof**

If $A$ is a field, then let $0 \neq b \in B$. It is integral over $A$ so $b^n + a_1 b^{n-1} + \cdots + a_n = 0$ for $a_i \in A$, and assume $n$ is minimal. Then

$$b(b^{n-1} + a_1 b^{n-2} + \cdots + a_{n-1}) = -a_n.$$

Since by minimality $b^{n-1} + a_1 b^{n-1} + \cdots + a_{n-1} \neq 0$, we have $a_n \neq 0$. Thus $-a_n$ is a unit, and thus $b$ must be a unit as well (if $xy = u$ is a unit, then $x$ is a unit since $x(yu^{-1}) = 1$). So $B$ is a field.

Conversely if $B$ is a field, then let $0 \neq x \in A$. Then $x^{-1} \in B$ since it is a field. Since $B$ is integral over $A$, there are $a_i \in A$ such that

$$x^{-n} + a_1 x^{-(n-1)} + \cdots + a_n = 0,$$

and multiplying by $x^{n-1}$ we get

$$x^{-1} + a_1 + \cdots + a_n x^{n-1},$$

so that $x^{-1}$ is the sum of elements from $A$, and thus is in $A$. ■

**1.6.20 Corollary**

Let $B$ be an integral $A$-algebra, and $\mathfrak{q} \leq B$ a prime ideal. Let $\mathfrak{p} = \mathfrak{q}_A \leq A$ be its preimage, then $\mathfrak{q}$ is maximal iff $\mathfrak{p}$ is maximal.

**Proof**

Both $\mathfrak{p}, \mathfrak{q}$ are prime so that $B/\mathfrak{q}$ and $A/\mathfrak{p}$ are integral domains. As we showed, $B/\mathfrak{q}$ is integral over $A/\mathfrak{p}$. Thus $B/\mathfrak{q}$ is a field iff $A/\mathfrak{p}$ is a field; so $\mathfrak{q}$ is maximal iff $\mathfrak{p}$ is maximal. ■

**1.6.21 Corollary**

Let $B$ be an integral $A$-algebra, $\mathfrak{q} \leq \mathfrak{q}' \leq B$ be prime ideals such that $\mathfrak{q}_A = \mathfrak{q}'_A$. Then $\mathfrak{q} = \mathfrak{q}'$.

**Proof**

Let $\mathfrak{p} = \mathfrak{q}_A = \mathfrak{q}'_A$; it is prime over $A$. Consider the following commutative square

$$\begin{tikzcd} A \arrow[r, "f"'] \arrow[d, "\iota_A"'] & B \arrow[d, "\iota_B"] \\ A_{\mathfrak{p}} \arrow[r, "f_{\mathfrak{p}}"] & B_{\mathfrak{p}} \end{tikzcd}$$

which commutes since the structure morphism $A \to S^{-1}A$ is a natural transformation (literally what the commutative square says).

Since $f^{-1}\mathfrak{q} = \mathfrak{p}$ we see that $\mathfrak{q} \cap f(A - \mathfrak{p}) = \emptyset$ and similarly for $\mathfrak{q}'$. Thus $\mathfrak{q}B_{\mathfrak{p}} \subseteq \mathfrak{q}'B_{\mathfrak{p}} \subseteq B_{\mathfrak{p}}$ are prime ideals (since they don't intersect the localizing set $f(A - \mathfrak{p})$).

We are going to show that $\mathfrak{q}B_{\mathfrak{p}} \subseteq B_{\mathfrak{p}}$ is a maximal ideal so that $\mathfrak{q}B_{\mathfrak{p}} = \mathfrak{q}'B_{\mathfrak{p}}$. And then since $\mathfrak{q} \mapsto \mathfrak{q}B_{\mathfrak{p}}$ is bijective with inverse $\iota_B^{-1}$, we will have that $\mathfrak{q} = \mathfrak{q}'$.

Since $B$ is integral over $A$, $B_\mathfrak{p}$ is integral over $A_\mathfrak{p}$. Thus, by the above corollary, it is sufficient to show that $\tilde{\mathfrak{p}} = f_\mathfrak{p}^{-1}(\mathfrak{q}B_\mathfrak{p}) \subseteq A_\mathfrak{p}$ is maximal. By assumption,

$$\iota_A^{-1}(\tilde{\mathfrak{p}}) = \iota_A^{-1}(f_\mathfrak{p}^{-1}(\mathfrak{q}B_\mathfrak{p})) = f^{-1} \circ \iota_B^{-1}(\mathfrak{q}B_\mathfrak{p}) = f^{-1}(\mathfrak{q}) = \mathfrak{p}.$$

Thus we have that $\tilde{\mathfrak{p}} = \mathfrak{p}A_\mathfrak{p}$ (since $\iota_A^{-1}$ is bijective) which is maximal, as desired. ■

The main result we want is the following.

**1.6.22 Theorem**

Let $f\colon A \to B$ be an injective integral morphism. Then for every prime $\mathfrak{p} \leq A$, there exists a prime ideal $\mathfrak{q} \leq B$ such that $\mathfrak{p} = \mathfrak{q}_A$.

**Proof**

Since $B_\mathfrak{p} = f(A - \mathfrak{p})^{-1}B$, and $0 \notin f(A - \mathfrak{p})$ because $f$ is injective, $B_\mathfrak{p} \neq 0$. As before, $B_\mathfrak{p}$ is integral over $A_\mathfrak{p}$ and we have the same commutative square

$$\begin{tikzcd} A \arrow[r, "f"'] \arrow[d, "\iota_A"'] & B \arrow[d, "\iota_B"] \\ A_\mathfrak{p} \arrow[r, "f_\mathfrak{p}"] & B_\mathfrak{p} \end{tikzcd}$$

Let $\mathfrak{m} \subseteq B_\mathfrak{p}$ be maximal, so that $f_\mathfrak{p}^{-1}(\mathfrak{m}) \subseteq A_\mathfrak{p}$ is maximal (as we showed $\mathfrak{q}_A$ is maximal iff $\mathfrak{q}$ is). Thus $f_\mathfrak{p}^{-1}(\mathfrak{m}) = \mathfrak{p}A_\mathfrak{p}$. Let $\mathfrak{q} = \iota_B^{-1}(\mathfrak{m})$, which is prime in $B$, and

$$f^{-1}(\mathfrak{q}) = f^{-1} \circ \iota_B^{-1}(\mathfrak{m}) = \iota_A^{-1} \circ f_\mathfrak{p}^{-1}(\mathfrak{m}) = \iota_A^{-1}(\mathfrak{p}A_\mathfrak{p}) = \mathfrak{p},$$

as desired. ■

**Note:**

Equivalently, this theorem says that if $f\colon A \to B$ is integral injective, then $f^*\colon \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is surjective.

Note that injectivity is necessary here. This is because $\pi\colon A \to A/I$ is integral for any ideal $I \leq A$, but $\pi^*\colon \operatorname{Spec}(A/I) \to \operatorname{Spec}(A)$ is not surjective: its image is $V(I)$, the set of prime ideals containing $I$.

**1.6.23 Corollary**

Let $B$ be an integral $A$-algebra, then for every prime $\mathfrak{q} \leq B$ and every prime ideal $\mathfrak{p}' \supseteq \mathfrak{p} = \mathfrak{q}_A$, there exists a prime ideal $\mathfrak{q}' \supseteq \mathfrak{q}$ of $B$ such that $\mathfrak{q}'_A = \mathfrak{p}'$.

**Proof**

Consider the commutative square

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\pi_A"'] & B \arrow[d, "\pi_B"] \\ A/\mathfrak{p} \arrow[r, "\overline{f}"] & B/\mathfrak{q} \end{tikzcd}$$

Then $\mathfrak{p}' \supseteq \mathfrak{p}$ gives rise to a prime ideal $\overline{\mathfrak{p}}' = \mathfrak{p}'/\mathfrak{p} \subseteq A/\mathfrak{p}$. Since $B/\mathfrak{q}$ is an integral $A/\mathfrak{p}$-algebra, there is a prime ideal $\overline{\mathfrak{q}}' \subseteq B/\mathfrak{q}$ such that $\overline{f}^{-1}(\overline{\mathfrak{q}}') = \overline{\mathfrak{p}}'$. Then $\mathfrak{q}' = \pi_B^{-1}(\overline{\mathfrak{q}}')$ is a prime ideal of $B$ containing $\mathfrak{q}$ and

$$f^{-1}(\mathfrak{q}') = f^{-1} \circ \pi_B^{-1}(\overline{\mathfrak{q}}') = \pi_A^{-1} \circ \overline{f}^{-1}(\overline{\mathfrak{q}}') = \pi_A^{-1}(\overline{\mathfrak{p}}') = \mathfrak{p}',$$

as desired. ■

**1.6.24 Proposition**

Let $B$ be an $A$-algebra, $C \subseteq B$ be the integral closure of $A$ inside of $B$. Then for any multiplicative $S \subseteq A$, $S^{-1}C$ is the integral closure of $S^{-1}A$ inside of $S^{-1}B$.

**Proof**

Since $C$ is integral over $A$, $S^{-1}C$ is integral over $S^{-1}C$. Now if $b/s$ is integral over $S^{-1}A$, then $b/1 = s \cdot b/s$ is integral over $S^{-1}A$ as well, and thus

$$\left(\frac{b}{1}\right)^n + \frac{a_1}{s_1}\left(\frac{b}{1}\right)^{n-1} + \cdots + \frac{a_n}{s_n} = 0$$

for $a_i/s_i \in S^{-1}A$. Let $t = s_1 \cdots s_n$ and $u_i = t/s_i \in S$, and multiplying this equation by $t^n$ gives us

$$\left(\frac{tb}{1}\right)^n + a_1u_1\left(\frac{bt}{1}\right)^{n-1} + \cdots + a_nu_nt^{n-1} = 0.$$

Thus there is a $v \in S$ such that

$$v((bt)^n + a_1u_1(bt)^{n-1} + a_2u_2t(bt)^{n-2} + \cdots + a_nu_nt^{n-1}) = 0.$$

Then $btv$ is integral over $A$ (multiply by $v^{n-1}$), so that $btv \in C$. Then $b/s = (btv)/(stv) \in S^{-1}C$, as desired. ■

**1.6.25 Corollary**

Let $A$ be a domain, then the following are equivalent:

**(1)** $A$ is normal,

**(2)** $S^{-1}A$ is normal for every multiplicative $S \subseteq A$ not containing 0,

**(3)** $A_\mathfrak{p}$ is normal for every prime $\mathfrak{p}$,

**(4)** $A_\mathfrak{m}$ is normal for every maximal $\mathfrak{m}$.

**Proof**

Let $K$ be $A$'s field of fractions, and let $C \subseteq K$ be $A$'s integral closure in $K$. Then $S^{-1}C$ is $S^{-1}A$'s integral closure in $S^{-1}K = K$ for all multiplicative $S$ not containing 0. If $A$ is normal, then $C = A$ so that $S^{-1}A$ is integrally closed in $K$, and thus normal. So (1) implies (2), which implies (3) which implies (4).

Now if $A_{\mathfrak{m}}$ is normal, then $A_{\mathfrak{m}} \to C_{\mathfrak{m}}$ is an isomorphism for all $\mathfrak{m}$. We showed that this implies $A \to C$ is an isomorphism, so that $A$ is normal. ■

# 2 Schemes

## 2.1 Affine schemes

**2.1.1 Definition**

Let $R$ be a ring, then its **(prime) spectrum** is its set of prime ideals,

$$\operatorname{Spec}(R) = \{\mathfrak{p} \leq R \mid \mathfrak{p} \text{ is prime}\}.$$

**Note:**

We can similarly define the **maximal spectrum** of a ring to be its set of maximal ideals. This is in general less useful than the prime spectrum, but its definition is analogous:

$$\operatorname{Specm}(R) = \{\mathfrak{m} \leq R \mid \mathfrak{m} \text{ is maximal}\}.$$

We can give $\operatorname{Spec}(R)$ a topological structure. Instead of defining the open sets of a spectrum, we will define its closed sets.

**2.1.2 Definition**

Let $S \subseteq R$ be a subset, then define its **vanishing set** to be

$$V(S) = \{\mathfrak{p} \in \operatorname{Spec}(R) \mid S \subseteq \mathfrak{p}\}.$$

Notice that clearly we have

$$V(S) = V((S)),$$

so that every vanishing set is a vanishing set of an ideal. Note that $V(I) = V(J)$ does not imply $I = J$: for instance $I = 2\mathbb{Z}$ and $J = 4\mathbb{Z}$ in $R = \mathbb{Z}$; their vanishing sets are both $\{2\mathbb{Z}\}$.

**2.1.3 Lemma**

A proper ideal $\mathfrak{p} \leq R$ is prime iff $IJ \leq \mathfrak{p}$ implies $I \leq \mathfrak{p}$ or $J \leq \mathfrak{p}$ for all ideals $I, J \leq R$.

**Proof**

The "if" part is clear: if $ab \in \mathfrak{p}$, then $(a) \cdot (b) = (ab) \in \mathfrak{p}$ so either $(a)$ or $(b)$ is a subset of $\mathfrak{p}$. Conversely, suppose that $\mathfrak{p}$ is a prime ideal and $IJ \leq \mathfrak{p}$ but $I, J$ are both not subsets of $\mathfrak{p}$. Then let $x \in I, y \in J$ be elements not in $\mathfrak{p}$: we then have $xy \in IJ \leq \mathfrak{p}$, a contradiction to $\mathfrak{p}$'s primality. ■

**2.1.4 Proposition**

The collection of vanishing sets forms the collection of closed sets of a topology on $\operatorname{Spec}(R)$.

**Proof**

We need to show that $\operatorname{Spec}(R), \emptyset$ are vanishing sets, the arbitrary intersection of vanishing sets is a vanishing set, and the binary union of vanishing sets is a vanishing set. Clearly we have

$$\operatorname{Spec}(R) = V(0), \qquad \emptyset = V(R)$$

so the first point is true.

Now consider a collection $\{S_\alpha\}_\alpha$ of subsets. Then we have:

$$\bigcap_\alpha V(S_\alpha) = V\left(\bigcup_\alpha S_\alpha\right),$$

since $S_\alpha \leq \mathfrak{p}$ for all $\alpha$ iff $\bigcup_\alpha S_\alpha \leq \mathfrak{p}$. In terms of ideals, we have that

$$\bigcap_\alpha V(I_\alpha) = V\left(\left(\bigcup_\alpha I_\alpha\right)\right) = V\left(\sum_\alpha I_\alpha\right).$$

For the final point on binary unions, we can restrict our attention to vanishing sets of ideals. Due to the above lemma we have that

$$V(I) \cup V(J) = V(IJ).$$

Thus the collection of vanishing sets forms a topology of closed sets. ■

**2.1.5 Definition**

The topology defined above on $\operatorname{Spec}(R)$ is called the **Zariski topology** of $\operatorname{Spec}(R)$.

**2.1.6 Example**

Consider $R = \mathbb{C}[x]$, a PID. Its prime ideals are the principal ideals generated by irreducible elements and zero, in this case

$$\operatorname{Spec}(\mathbb{C}[x]) = \{(x - \lambda) \mid \lambda \in \mathbb{C}\} \cup \{(0)\}.$$

For a polynomial $f \in \mathbb{C}[x]$, we have that $(x - \lambda) \in V(f)$ iff $\lambda$ is a root of $f$. Thus

$$V(f) = \{(x - \lambda) \mid f(\lambda) = 0\}.$$

As we showed above, for $I \leq \mathbb{C}[x]$, we have that

$$V(I) = V\left(\bigcup_{f \in I} \{f\}\right) = \bigcap_{f \in I} V(f) = \{(x - \lambda) \mid \lambda \text{ is a root of every polynomial in } I\}.$$

Thus $\operatorname{Spec}(\mathbb{C}[x])$'s proper closed sets are precisely the finite subsets of $\operatorname{Spec}(\mathbb{C}[x]) - (0)$, i.e. it induces the cofinite topology on this subspace. The ideal $(0)$ is not contained in any proper closed set, so its closure is all of $\operatorname{Spec}(\mathbb{C}[x])$.

**2.1.7 Example**

Consider $R = \mathbb{Z}$, a PID. Its prime ideals are the principal ideals generated by primes and zero, so

$$\operatorname{Spec}(\mathbb{Z}) = \{(p) \mid p \text{ prime}\} \cup \{(0)\}.$$

For $n \in \mathbb{Z}$, $(p) \in V(n)$ iff $p$ divides $n$ so that

$$V(n) = \{(p) \mid p \text{ is a prime divisor of } n\}.$$

Thus $\operatorname{Spec}(\mathbb{Z})$'s proper closed sets are precisely the finite subsets of $\operatorname{Spec}(\mathbb{Z}) - 0$, so it again induces the cofinite topology on this subspace. As before, $(0)$'s closure is all of $\operatorname{Spec}(\mathbb{Z})$.

In the two examples above, we saw that the point $(0)$'s closure is all of $\operatorname{Spec}(R)$. If $(0)$ is a prime ideal, then this is always true: $(0) \in V(I)$ iff $I = (0)$ in which case $V(0) = \operatorname{Spec}(R)$. Thus $\operatorname{Spec}(R)$ is the only closed set containing the zero ideal, and is thus its closure.

**2.1.8 Example**

Consider the case that $R = k$ is a field. The only prime ideal of $k$ is the zero ideal, so $\operatorname{Spec}(k)$ is a single-point space. There is only one topology on such a space: the trivial and discrete topology coincide.

The map $R \mapsto \operatorname{Spec}(R)$ is functorial.

**2.1.9 Definition**

Let $\phi\colon R \to S$ be a ring morphism, then define $\phi^*\colon \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ by

$$\phi^*(\mathfrak{q}) = \phi^{-1}(\mathfrak{q}), \qquad \mathfrak{q} \in \operatorname{Spec}(S).$$

This is well-defined, as we showed before that the preimage of a prime ideal is prime.

**2.1.10 Proposition**

$\phi^*$ is continuous, and functorial.

**Proof**

To show that $\phi^*$ is continuous, we need to show that its preimage on closed sets is closed. So let $I \leq R$ be an ideal, and consider $(\phi^*)^{-1}V(I)$:

$$\mathfrak{q} \in (\phi^*)^{-1}V(I) \iff \phi^{-1}\mathfrak{q} \in V(I) \iff I \leq \phi^{-1}\mathfrak{q} \iff \phi(I) \leq \mathfrak{q},$$

so that we have

$$(\phi^*)^{-1}V(I) = V(\phi(I)),$$

which is closed as desired. Note that while $\phi(I) \subseteq S$ may not be an ideal, all vanishing sets are closed. Functorality is clear. ■

**Note:**

As $\mathrm{Specm}(R) \subseteq \mathrm{Spec}(R)$, we endow $\mathrm{Specm}(R)$ with the subspace topology. In particular its closed sets are

$$V_m(S) = \{\mathfrak{m} \in \mathrm{Specm}(R) \mid S \subseteq \mathfrak{m}\}.$$

But note that $R \mapsto \mathrm{Specm}(R)$ is not functorial: the above $\phi^*$ does not necessarily restrict to maximal ideals.

Let us consider more the Zariski topology. Clearly $V(\bullet)$ is contravariant (order reversing).

**2.1.11 Proposition**

Let $I, J \leq R$ be ideals, then $V(I) = V(J)$ iff $\sqrt{I} = \sqrt{J}$. In particular $V(I) = V(\sqrt{I})$ and $\mathrm{Spec}(R) = V(\mathrm{nil}(R))$.

**Proof**

We know that $\sqrt{I}$ is the intersection of all prime ideals containing $I$, meaning

$$\sqrt{I} = \bigcap_{\mathfrak{p} \in V(I)} \mathfrak{p}.$$

So the first implication is clear: if $V(I) = V(J)$ then $\sqrt{I} = \sqrt{J}$. Now if $\sqrt{I} = \sqrt{J}$, let $\mathfrak{p} \in V(I)$. If $\mathfrak{p} \notin V(J)$ then there is an $f \in J$ not in $\mathfrak{p}$, but then $f \in \sqrt{J}$ and $f \notin \mathfrak{p}$ so $f \notin \sqrt{I}$, a contradiction. ■

**Note:**
Let the **Jacobson radical** of an ideal $I \leq R$ be the intersection of all maximal ideals containing $I$:

$$\mathrm{Jac}_R(I) = \bigcap_{I \leq \mathfrak{m}} \mathfrak{m} = \bigcap_{\mathfrak{m} \in V_m(I)} \mathfrak{m}.$$

We can alter the proof above to show that $V_m(I) = V_m(J)$ iff $\mathrm{Jac}_R(I) = \mathrm{Jac}_R(J)$. Thus $V_m(I) = V_m(\mathrm{Jac}_R(I))$ and $\mathrm{Specm}(R) = V_m(\mathrm{Jac}(R))$, where $\mathrm{Jac}(R) = \mathrm{Jac}_R(R)$ is the usual Jacobson radical of a ring.

**2.1.12 Definition**

Let $Z \subseteq \mathrm{Spec}(R)$ be a subset. Define its **ideal** to be

$$\mathcal{I}(Z) = \{f \in R \mid Z \subseteq V(f)\}.$$

We can characterize $\mathcal{I}(Z)$ more concretely:

$$\mathcal{I}(Z) = \bigcap_{\mathfrak{p} \in Z} \mathfrak{p}.$$

Indeed, if $f \in \mathcal{I}(Z)$ then $Z \subseteq V(f)$ so for every $\mathfrak{p} \in Z$ we have that $\mathfrak{p} \in V(f)$, meaning $f \in \mathfrak{p}$. Conversely if $f \in \mathfrak{p}$ for every $\mathfrak{p} \in Z$, then $\mathfrak{p} \in V(f)$ for every $\mathfrak{p}$ and so $Z \subseteq V(f)$.
From this characterization it is obvious that $\mathcal{I}(\bullet)$ is also order reversing.

**2.1.13 Proposition**

**(1)** Let $Z \subseteq \operatorname{Spec}(R)$ be a subset, then $V(\mathcal{I}(Z)) = \overline{Z}$, $Z$'s closure.

**(2)** Let $J \leq R$ be an ideal, then $\mathcal{I}(V(J)) = \sqrt{J}$.

**Proof**

For the first point we need to show that $V(\mathcal{I}(Z))$ is the smallest vanishing set containing $Z$. It contains $Z$ since if $\mathfrak{p} \in Z$ then $\mathcal{I}(Z) \leq \mathfrak{p}$ by definition (it is the intersection of all such $\mathfrak{p}$), so $\mathfrak{p} \in V(\mathcal{I}(Z))$. Now if $Z \subseteq V(J)$ then for $\mathfrak{p} \in Z$ we have that $J \leq \mathfrak{p}$, so that $J \subseteq \bigcap_{\mathfrak{p} \in Z} \mathfrak{p} = \mathcal{I}(Z)$. Since $V$ is contravariant, we have that $V(\mathcal{I}(Z)) \subseteq V(J)$, as desired.

For the second point, we have

$$\mathcal{I}(V(J)) = \bigcap_{\mathfrak{p} \in V(J)} \mathfrak{p} = \bigcap_{J \leq \mathfrak{p}} \mathfrak{p} = \sqrt{J},$$

as desired. ■

**2.1.14 Corollary**

Let $\mathfrak{p} \in \operatorname{Spec}(R)$ be a point, then its closure is $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$. In particular $\{\mathfrak{p}\}$ is closed iff $\mathfrak{p}$ is maximal.

**Proof**

This is because $\mathcal{I}\{\mathfrak{p}\} = \mathfrak{p}$, so apply the above proposition. And $V(\mathfrak{p}) = \{\mathfrak{p}\}$ iff $\mathfrak{p}$ is maximal. ■

**2.1.15 Example**

This shows that $\operatorname{Spec}(R)$ need not be $T_1$. Recall that a space $X$ is $T_1$ if for every two distinct $x \neq y \in X$, there is an open set $U$ such that $x \in U$ and $y \notin U$. Equivalently a space is $T_1$ iff its singletons are closed: if singletons are closed let $U = \{y\}^c$, and conversely if for every $y \neq x$ we choose a $U_y$ such that $y \in U_y$ and $x \notin U_y$ we see that

$$X - \{x\} = \bigcup_{y \neq x} U_y$$

is open and thus $\{x\}$ is closed.

In any case, clearly if we have prime ideals which are not maximal, then $\operatorname{Spec}(R)$ is not $T_1$. For instance, take $X = \operatorname{Spec}(\mathbb{Z})$. Since $(0) \in X$ is not maximal, it is not closed and thus the space cannot be $T_1$.

$\operatorname{Spec}(R)$ *is* $T_0$ though: all distinct points are topologically distinct. Indeed, if $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec}(R)$ have the same neighborhoods they have the same closure and thus $V(\mathfrak{p}) = V(\mathfrak{q})$. Thus $\mathfrak{p} \in V(\mathfrak{q})$ and $\mathfrak{q} \in V(\mathfrak{p})$ so that $\mathfrak{p} \leq \mathfrak{q} \leq \mathfrak{p}$ so that they are equal.

**2.1.16 Definition**

Let $P, Q$ be two posets. An **antitone Galois connection** are two order-reversing functions $F\colon P \to Q$ and $G\colon Q \to P$ such that for every $p \in P, q \in Q$,

$$q \leq Fp \iff p \leq Gq.$$

An alternative method of looking at antitone Galois connections is via category theory. Recall that we can view posets as categories, where there exists a unique morphism $a \to b$ iff $a \leq b$.

**2.1.17 Proposition**

An antitone Galois connection between two posets $P, Q$ consists of two functors $F\colon P \to Q^{\mathsf{op}}\colon G$ such that $F \dashv G$ forms an adjunction.

**Proof**

Order reversing functions between posets are just contravariant functors. So if $F, G$ form an antitone Galois connection, then they are functors $F\colon P^{\mathsf{op}} \to Q$ and $G\colon Q^{\mathsf{op}} \to P$. Then $F^{\mathsf{op}}\colon P \to Q^{\mathsf{op}}\colon G$ forms an adjunction: we have $q \leq Fp$ iff $F^{\mathsf{op}}p \leq^{\mathsf{op}} q$ iff $p \leq Gq$. The converse is shown similarly. ■

**2.1.18 Corollary**

$F\colon P \to Q\colon G$ forms an antitone Galois connection iff they are order reversing and $p \leq GFp$ and $q \leq FGq$ for all $p \in P, q \in Q$.

**Proof**

This is just the unit-counit formulation of an adjunction (remember contravariance!). A natural transformation between functors $X, Y$ between posets amounts to saying $X(\bullet) \leq Y(\bullet)$. ■

**2.1.19 Corollary**

The two functions $V\colon \mathcal{P}(R) \to \mathcal{P}(\mathrm{Spec}(R))\colon \mathcal{I}$ form an antitone Galois connection.

**Proof**

The maps are order reversing and $S \subseteq \sqrt{(S)} = \mathcal{I}(V(S))$ and $Z \subseteq \overline{Z} = V(\mathcal{I}(Z))$, so they form an antitone Galois connection. Showing directly that $Z \subseteq V(S)$ iff $S \subseteq \mathcal{I}(Z)$ is easy to do as well. ■

**2.1.20 Definition**

Let $R$ be a ring, $X = \mathrm{Spec}(R)$ its spectrum. For $f \in R$, define

$$X_f = X - V(f) = \{\mathfrak{p} \in \mathrm{Spec}(R) \mid f \notin \mathfrak{p}\}\,.$$

**2.1.21 Proposition**

The collection $\{X_f\}_{f \in R}$ forms a basis of open sets for $X = \mathrm{Spec}(R)$.

**Proof**

This is equivalent to showing that $\{V(f)\}_{f \in R}$ forms a basis of closed sets; that they are closed and every

closed set is an intersection of closed basis sets. Clearly $V(f)$ are closed, and for $S \subseteq R$ we have

$$V(S) = V\left(\bigcup_{f \in S} \{f\}\right) = \bigcap_{f \in S} V(f)$$

as desired. ■

This shows that for an open set $X - V(S)$, we have

$$X - V(S) = \bigcup_{f \in S} X_f.$$

**2.1.22 Proposition**

Let $R$ be a ring and $X = \operatorname{Spec}(R)$.

**(1)** Let $\{f_\alpha\}_\alpha$ be a collection of elements from $R$. Then $X = \bigcup_\alpha X_{f_\alpha}$ iff $(\{f_\alpha\}_\alpha) = R$.

**(2)** Let $f \in R$, then $X_f = X$ iff $f \in R^\times$.

**(3)** Let $f \in R$, then $X_f = \emptyset$ iff $f \in \operatorname{nil}(R)$.

**Proof**

We saw that

$$\bigcup_\alpha X_{f_\alpha} = X - V(\{f_\alpha\}_\alpha)$$

which is equal to $X$ iff $V(\{f_\alpha\}_\alpha) = \emptyset$. This is iff

$$V(\{f_\alpha\}_\alpha) = V((\{f_\alpha\}_\alpha)) = V(R) \iff \sqrt{(\{f_\alpha\}_\alpha} = R.$$

Now $\sqrt{J} = R$ iff $J = R$, as $1 \in \sqrt{J}$ iff $1 \in J$. So this is iff $(\{f_\alpha\}_\alpha) = R$, as desired.
The second point follows from the first: $X_f = X$ iff $(f) = R$ iff $f \in R^\times$.
For the third point, $X_f = \emptyset$ iff $V(f) = X = V(0)$ iff $\sqrt{(f)} = \operatorname{nil}(R)$. This occurs iff $f \in \operatorname{nil}(R)$: clearly it implies this, and conversely $0 \leq (f) \leq \operatorname{nil}(R)$ and radicals are order-preserving so we get $\sqrt{(f)} = \operatorname{nil}(R)$. ■

**2.1.23 Corollary**

$X = \operatorname{Spec}(R)$ is compact.

**Proof**

We need to show every open cover of $X$ has a finite subcover. Let $\mathcal{U}$ be an open cover of $X$, then we can assume that $\mathcal{U}$ consists of basic open sets: $\mathcal{U} = \{X_{f_\alpha}\}_\alpha$. By the first point above, since $\mathcal{U}$ is a cover, $(\{f_\alpha\}_\alpha) = R$. Thus $1 \in (\{f_\alpha\}_\alpha)$ so there are finitely many $f_1, \ldots, f_n$ such that $1 = x_1 f_1 + \cdots + x_n f_n$. Then $(f_1, \ldots, f_n) = R$ and so again by the above proposition, $\{X_{f_1}, \ldots, X_{f_n}\}$ is a finite subcover of $\mathcal{U}$, as desired. ■

**Note:**
This result shows that the usual topological concept of compactness is in general not strong enough for algebraic geometry. Thus in algebraic geometry topological compactness is usually called **quasicompactness**, and compactness is reserved for a stronger concept.

**2.1.24 Proposition**

Let $f \in R$, and consider $R_f$, the localization of $R$ at $\{f^n\}_{n<\omega}$. $\mathrm{Spec}(R_f)$ is homeomorphic to $X_f$.

**Proof**

Let $\iota: R \to R_f$ be the structure map. Then $\mathfrak{q} \mapsto \iota^{-1}\mathfrak{q}$ is a bijective correspondence between prime ideals of $R_f$ and prime ideals of $R$ which do not intersect $\{f^n\}$. A prime ideal $\mathfrak{p} \leq R$ does not intersect $\{f^n\}$ iff $f \notin \mathfrak{p}$. Thus $\iota^*: Y = \mathrm{Spec}(R_f) \to \mathrm{Spec}(R)$ is injective and its image are all prime ideals such that $f \notin \mathfrak{p}$, i.e. $\mathrm{im}\iota^* = X_f$.

We claim that $\iota^*$ is a homeomorphism onto its image. We already know it is continuous and bijective, so we need only show it is open; we will show it sends basic open sets to basic open sets. So let $g/f^n \in R_f$, with the basic open set

$$Y_{g/f^n} = \{\mathfrak{q} \in Y \mid g/f^n \notin \mathfrak{q}\}.$$

We claim that $\iota^*(Y_{g/f^n}) = X_g \cap X_f$. Indeed, if $\mathfrak{q} \in Y_{g/f^n}$ then $g/f^n \notin \mathfrak{q}$ and so $g \notin \iota^*\mathfrak{q}$ (since $g/f^n$ is $\iota(g)$ times a unit, so if $g \in \iota^*\mathfrak{q}$ then $\iota(g) \in \mathfrak{q}$ and then $g/f^n \in \mathfrak{q}$); thus $\iota^*\mathfrak{q} \in X_g$ as desired. Conversely if $\mathfrak{p} \in X_f \cap X_g$ then $g/f^n \notin \mathfrak{p}R_f$: otherwise $g/f^n = h/f^m$ for $h \in \mathfrak{p}$ and thus $f^k(f^m g - f^n h) = 0$ so that $f^{m+k}g \in \mathfrak{p}$, but $f, g \notin \mathfrak{p}$ a contradiction. Now, $\mathfrak{p}R_f$ is prime and $\iota^*\mathfrak{p}R_f = \mathfrak{p}$, showing the equality. ■

We now focus on the connectedness of topological spaces, in particular spectra of rings. Recall that a topological space is **disconnected** if it can be written as the disjoint union of two open subsets, equivalently if it has a non-trivial clopen subset.

**2.1.25 Lemma**

A surjective ring morphism $\phi: R \to S$ induces a closed embedding $\phi^*: \mathrm{Spec}(S) \to \mathrm{Spec}(R)$ (i.e. its image is closed, and it is a topological embedding). In particular if $I \leq R$ is an ideal, then the projection $\pi: R \to R/I$ induces a closed embedding $\pi^*: \mathrm{Spec}(R/I) \to \mathrm{Spec}(R)$ whose image is $V(I)$.

**Proof**

$\phi$ induces an isomorphism of rings $R/\ker\phi \to S$, which carries prime ideals of $R/\ker\phi$ bijectively to prime ideals of $S$. Thus $\phi^*$'s image is the set of all prime ideals containing $\ker\phi$, i.e. $\mathrm{im}\phi^* = V(\ker\phi)$ which is closed by definition. Since $\phi$ is surjective, $\phi^*$ is injective and thus a bijection onto its image. So to show it is a topological embedding, we need only show that it is a closed map.

Indeed, for $J \leq S$,

$$\phi^* V(J) = \left\{\phi^{-1}\mathfrak{q} \mid J \leq \mathfrak{q}\right\} = \left\{\phi^{-1}\mathfrak{q} \mid \phi^{-1}J \leq \phi^{-1}\mathfrak{q}\right\} = V(\phi^{-1}J) \cap \mathrm{im}\phi^*,$$

which is closed. (The second equality is due to $\phi$'s surjectivity.) ■

**2.1.26 Proposition**

If $R = S \times T$ is a product ring, then $\mathrm{Spec}(R)$ is homeomorphic to $\mathrm{Spec}(S) \sqcup \mathrm{Spec}(T)$, and thus disconnected.

Conversely if $\operatorname{Spec}(R) = Y \sqcup Z$ is disconnected, then there exist rings $S, T$ such that $Y, Z$ are homeomorphic to $\operatorname{Spec}(S), \operatorname{Spec}(T)$ respectively, and $R \cong S \times T$.

**Proof**

Let $X = \operatorname{Spec}(R)$. Consider $(1,0),(0,1) \in R$, so that $R$'s unit is their sum. Then we showed above that $X = X_{(1,0)} \cup X_{(0,1)}$, and this is a disjoint union since $V(0,1) \cup V(1,0) = X$ (if $\mathfrak{p} \in X$ and $(0,1),(1,0) \notin \mathfrak{p}$ then their product 0 is not in $\mathfrak{p}$, a contradiction). We also showed that $X_{(1,0)} \cong \operatorname{Spec}(R_{(1,0)})$.

So let $\pi\colon R \to S$ be the projection, so $\pi(1,0) = 1 \in S^\times$ so that by the universal property of localization $\pi$ induces a map $\pi'\colon R_{(1,0)} \to S$ which is surjective since $\pi$ is. Now if $\pi'((a,b)/(1,0)) = \pi'((x,y)/(1,0))$ then $a = x$ so $(a,b)(1,0) = (x,y)(1,0)$ and thus $(a,b)/(1,0) = (x,y)/(1,0)$ showing that $\pi'$ is also injective. Thus $S \cong R_{(1,0)}$, and similarly $T \cong R_{(0,1)}$ so that we have desired result.

Since $Y, Z$ are closed, there are $I, J \leq R$ such that $Y = V(I)$ and $Z = V(J)$. Then let $S = R/I$ and $T = R/J$; we showed above that $\pi_1\colon R \to R/I$ induces a homeomorphism $\pi_1^*\colon \operatorname{Spec}(R/I) \to V(I) \subseteq \operatorname{Spec}(R)$ and similar for $R/J$. Now, $Y \cap Z = V(I) \cap V(J) = V(I+J)$ is empty so that $I + J = R$, and thus $I, J$ are coprime. By the chinese remainder theorem,

$$R \cong R/I \times R/J = S \times T,$$

as desired. ■

**2.1.27 Definition**

An element $f \in R$ is **idempotent** if $f^2 = f$. The **trivial idempotents** of $R$ are 0 and 1.

If $f \in R$ is idempotent, then $fR$ is a ring. Indeed, it is closed under addition and multiplication (it is the ideal generated by $f$), and $f$ is its multiplicative identity since it is idempotent.

**2.1.28 Proposition**

$R \cong S \times T$ for non-trivial rings $S, T$ iff $R$ has nontrivial idempotents.

**Proof**

If $R \cong S \times T$ then $(1,0),(0,1)$ are nontrivial idempotents. Conversely let $f \in R$ be a nontrivial idempotent, then define $S = fR$ and $T = (1-f)R$. Then $R \to S \times T$ given by $g \mapsto (fg, (1-f)g)$ is clearly an injective since if $fg = (1-f)g = 0$ then $g = 0$, and it is surjective since for $(fg, (1-f)h) \in S \times T$ then we want a $x \in R$ such that $fg = fx$ and $(1-f)h = (1-f)x$, so $(1-f)h = x - fg$ and thus $x = (1-f)h + fg$ gives the desired element. They are also non-trivial since $f \neq 0$. ■

**2.1.29 Definition**

A ring $R$ is **connected** if it has no nontrivial idempotents.

**2.1.30 Corollary**

$\operatorname{Spec}(R)$ is connected iff $R$ is.

**2.1.31 Definition**

A topological space $X$ is **reducible** if there exist two proper closed subsets $Z, W \subsetneq X$ such that $X = Z \cup W$. Otherwise $X$ is **irreducible**.

Note that irreducibility implies connectedness; if $X$ is disconnected then its union $Z \sqcup W$ of open sets are also closed. The converse is not true in general (take the axes of $\mathbb{R}^2$).

**2.1.32 Proposition**

$X = \operatorname{Spec}(R)$ is irreducible iff $\operatorname{nil}(R)$ is prime.

**Proof**

If $\operatorname{nil}(R)$ is not prime, then there are $f, g \notin \operatorname{nil}(R)$ such that $fg \in \operatorname{nil}(R)$. Let $Z = V(f)$ and $W = V(g)$, so that

$$Z \cup W = V(f) \cup V(g) = V(fg) = X,$$

where the second equality is because $V(f) \cup V(g) = V((f)(g)) = V(fg)$, and the final equality is because $V(fg) \supseteq V(\operatorname{nil}(R)) = X$.

Conversely if $X$ is reducible, then $X = V(I) \cup V(J)$ so that $V(IJ) = X$ for $V(I), V(J) \neq X$. But then $IJ \leq \operatorname{nil}(R)$ and $I$ and $J$ cannot be contained in $\operatorname{nil}(R)$, so $\operatorname{nil}(R)$ is not prime. ■

**2.1.33 Corollary**

The spectrum of a domain is irreducible.

**2.1.34 Example**

The affine space $\mathbb{A}^n_{\mathbb{C}} = \operatorname{Spec}(\mathbb{C}[x_1, \ldots, x_n])$ is irreducible since $\mathbb{C}[x_1, \ldots, x_n]$ is a domain.

Similarly the spectrum of $R = \mathbb{C}[x_1, \ldots, x_n]/(f)$ is irreducible when $f$ is irreducible (and then $(f)$ is prime and $R$ is a domain). But the converse does not hold: let $R = \mathbb{C}[x]/(x^2)$, $x^2$ is reducible, and $\operatorname{nil}(R) = \sqrt{(x^2)} = (x)$ is prime.